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THE WHITE PINE COPPER DEPOSIT, ONTONAGON COUNTY, MICHIGAN¹

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CONTENTS

	PAGE
Abstract	676
Introduction	676
The Nonesuch shale	680
The cupriferous zone	680
Lower sandstone	681
Parting shale	682
Upper sandstone	683
Upper shale	683
Rocks above the cupriferous zone	684
Paleogeography of rocks of the cupriferous zone	685
Sandstones	685
Thinly laminated beds just above sandstones	687
Massive siltstone	687
Summary	688
Persistence of units away from White Pine	688
Structure	690
Faults	691
Folds	691
Structure near White Pine	691
Geology of copper minerals and related minerals	695
Relative abundance of minerals	695
Stratigraphy of copper distribution	696
Laminated bed, base of parting shale	702
Massive bed, middle parting shale	703
Laminated bed, base of upper shale	704
Copper in and above massive bed, upper shale	705
Copper in sandstone	706
Relation of copper to fissures and faults	708
Copper in fissures and joints	708
Relation of copper in shale to faults and fissures	709
Origin of copper	712
Copper in the Nonesuch shale in other areas	715
References	716

¹ Publication authorized by the Director, U. S. Geological Survey.

ABSTRACT

Copper, largely in the mineral chalcocite, is found in the lowermost beds of the Nonesuch shale over many square miles near White Pine, Ontonagon County, Mich. The Nonesuch shale, of late Keweenawan age, is about 600 feet thick and is composed largely of gray siltstone. It overlies 2,300 to 5,500 feet of red sandstones and conglomerates, the Copper Harbor conglomerate. This formation, in turn, overlies the middle Keweenawan Portage Lake lava series, in which occur the famous native-copper deposits of the south shore of Lake Superior. The White Pine mine is 45 to 70 miles west-southwest of the principal native copper mines.

The copper-bearing beds at White Pine are primarily in the lower 20 to 25 feet of the Nonesuch shale. This cupriferous zone is divided in local usage into four stratigraphic units, which are, in ascending order, the lower sandstone (the uppermost bed of the Copper Harbor conglomerate), the parting shale, the upper sandstone, and the upper shale. The sequence of beds is almost identical in the upper and parting shales, suggesting cyclic sedimentation. This sequence and the distribution of sedimentary facies are attributed to two submergences, separated by an emergence, of a deltaic area.

Practically all the copper occurs in the upper and parting shales, except in a small area near the White Pine fault where it is abundant in the upper and lower sandstones. It is present in five different layers in amounts that average from 1 to 3 percent. The total amount of copper in each bed is generally higher where the bed is thick, and thickness, in turn, seems to be greatest in areas that were hollows away from the main channel or channels of the ancient delta. Copper content of the shale beds typically decreases as their sand content increases. The extent of individual copper-bearing beds 1 to 3 feet thick is measurable in square miles.

The local occurrence of copper in the upper and lower sandstone beds can be reasonably explained as the result of hydrothermal transportation from the White Pine fault up the dip of the relatively permeable sandstone to the crest of an adjacent anticline. Distribution of copper in the parting and upper shales, on the other hand, seems to be completely independent of local structure, faults, and rock permeability. For the area as a whole the control is lithologic and stratigraphic. Probably the copper was dropped into or precipitated within the original mud; it may conceivably have replaced with exquisite detail something else that was so deposited.

INTRODUCTION

THE copper deposit at White Pine, Ontonagon County, Mich. (Fig. 1), is separated both geographically and stratigraphically from the famous native-copper deposits of the south shore of Lake Superior. The principal native-copper deposits lie 45 to 70 miles east-northeast of White Pine. Stratigraphically these native-copper deposits occur in the middle Keweenawan Portage Lake lava series (13),² whereas the deposit at White Pine lies at the base of the upper Keweenawan Nonesuch shale; the Copper Harbor conglomerate, 2,300 to 5,500 feet thick, separates the two formations.

The lowermost beds of the Nonesuch shale seem to be most cupriferous in the vicinity of the Porcupine Mountains (Fig. 1). Two mines, the Nonesuch ("H" in Fig. 1) and White Pine (1 mile northwest of "I," Fig. 1), intermittently exploited patches of high-grade ore from 1870 to 1920, and the formation

² Numbers in parentheses refer to References at end of paper.

was explored by drilling, test-pitting, or underground exploration at a number of other places. The Copper Range Co., through its subsidiary the White Pine Copper Co., is now undertaking to mine a multimillion-ton body of low-grade ore underlying an area of several square miles (parts of shaded area, Fig. 1) north and east of the old White Pine mine (2, 10).

The underground workings and diamond drill holes by which this large ore body has been developed provide a wealth of information on the geology of copper minerals in the Nonesuch shale. Several characteristics of the copper distribution indicate that the metal was deposited before deformation and probably before lithification of the rocks in which it now occurs. Locally there has been hydrothermal enrichment of sandstone beds, forming ore deposits like that at the old White Pine mine. This paper presents a preliminary geologic description of the cupriferous zone at the base of the Nonesuch shale and the distribution of copper minerals within it and gives the reasons for the conclusion that the copper deposit was present before lithification and deformation. This is part of a more detailed study of the Nonesuch shale that the authors are preparing.

The Nonesuch shale and its copper have been briefly described by Whittlesey (15), Rominger (11, 12), Irving (5), and Butler, Burbank, and others (3). Nishio (9) investigated the relation of carbonaceous matter to the native copper in the sandstone of the old White Pine mine. The most complete description of the occurrence of copper, largely the work of Dr. T. M. Broderick, is contained in U. S. Geological Survey Professional Paper 144 (3, p. 169-175). This study was based almost entirely on observations in the old White Pine mine. As will be apparent from the descriptions that follow, the character and distribution of copper in the old White Pine mine is closely related to structure and permeability and is very different from that in the low-grade deposit now being exploited. These differences in mode of occurrence are here attributed to differences in the geologic history of the copper.

The geologic data presented here have been collected intermittently over a period of six years as part of a geologic study of the Michigan Copper District by the U. S. Geological Survey. The senior author made a detailed study of the mile-long exploratory drift at the Schacht shaft ("I," Fig. 1) as it was being driven in 1946-47, with primary emphasis on a study of the faults and geologic structure. During the years 1949-52 the authors examined all the core preserved from the Copper Range Co. drilling and studied all the exposed sections and outcrops of the Nonesuch shale that they could find in the 125 miles between Calumet, Mich., and the Bad River north of Mellen, Wis. (Fig. 1).

This study could not have been undertaken without the whole-hearted cooperation of officials of the Copper Range Co. and the White Pine Copper Co. The authors are particularly grateful to Morris F. LaCroix, President, and Harold B. Ewoldt, Vice-president, to W. E. Romig, formerly General Manager, and to John R. Rand, Geologist, for unrestricted access to mine workings, drill core, and assay and other engineering records, and for many friendly courtesies throughout the work. They are also much indebted to Dr. T. M.

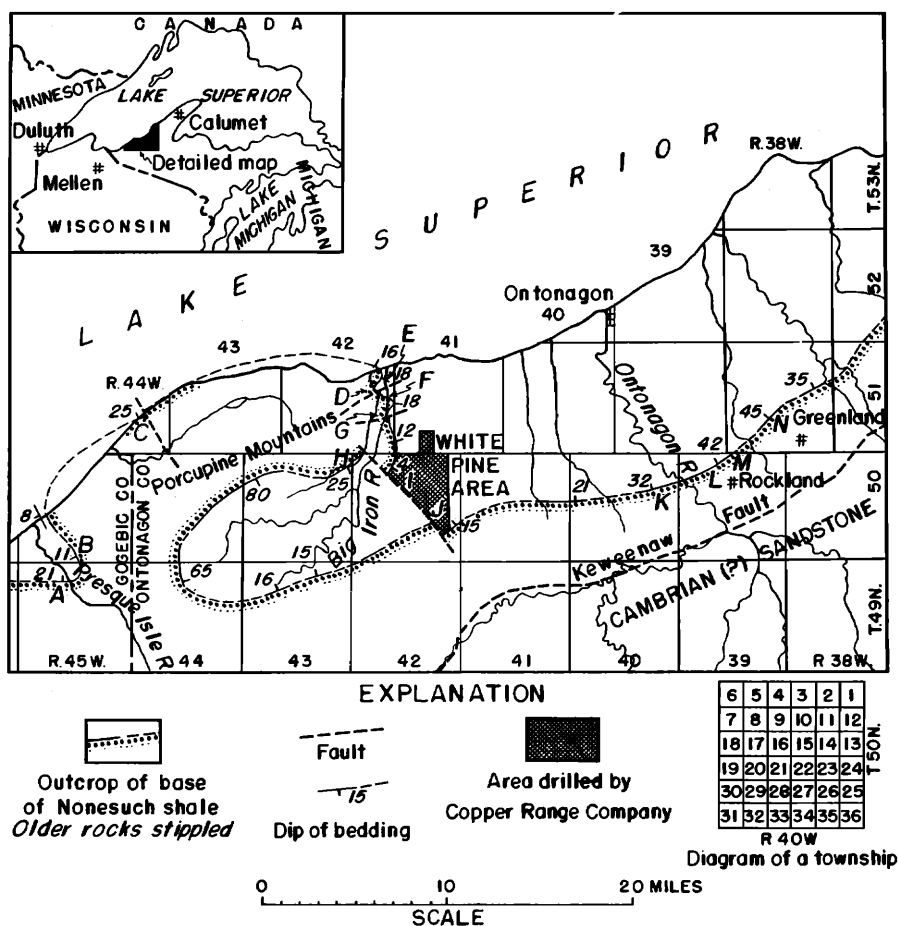


FIG. 1. Index map showing location of White Pine area and location and attitude of the base of the Nonesuch shale. Localities referred to in text and on Figure 6 indicated by letters as follows: A. Presque Isle River, SW $\frac{1}{4}$ sec. 4, T. 49 N., R. 45 W.; B. Tiebel Creek, north of the east quarter corner, sec. 33, T. 50 N., R. 45 W.; C. One mile southwest of Lone Rock, NW $\frac{1}{4}$ sec. 26, T. 51 N., R. 44 W.; D. Little Iron River, Scranton silver mine, SW $\frac{1}{4}$ sec. 13, T. 51 N., R. 42 W.; E. Big Iron River, below Greenwood camp, NE $\frac{1}{4}$ sec. 13, T. 51 N., R. 42 W.; F. Big Iron River, Superior silver mine, SE $\frac{1}{4}$ sec. 13, T. 51 N., R. 42 W.; G. Big Iron River, Collins silver mine, NE $\frac{1}{4}$ sec. 25, T. 51 N., R. 42 W.; H. Little Iron River, Nonesuch mine, SE $\frac{1}{4}$ sec. 1, T. 50 N., R. 43 W.; I. White Pine mine (Schacht shaft), see Fig. 7; J. White Pine drill hole 10-0, SE $\frac{1}{4}$ sec. 24, T. 50 N., R. 42 W.; K. Gates Creek, NW $\frac{1}{4}$ sec. 14, T. 50 N., R. 40 W.; L. Gravel pit near Rockland, SE $\frac{1}{4}$ sec. 5, T. 50 N., R. 39 W.; M. Rockland Creek, near center of the W $\frac{1}{2}$ sec. 4, T. 50 N., R. 39 W.; N. McGuire Creek, Greenland, NW $\frac{1}{4}$ sec. 28, T. 51 N., R. 38 W.

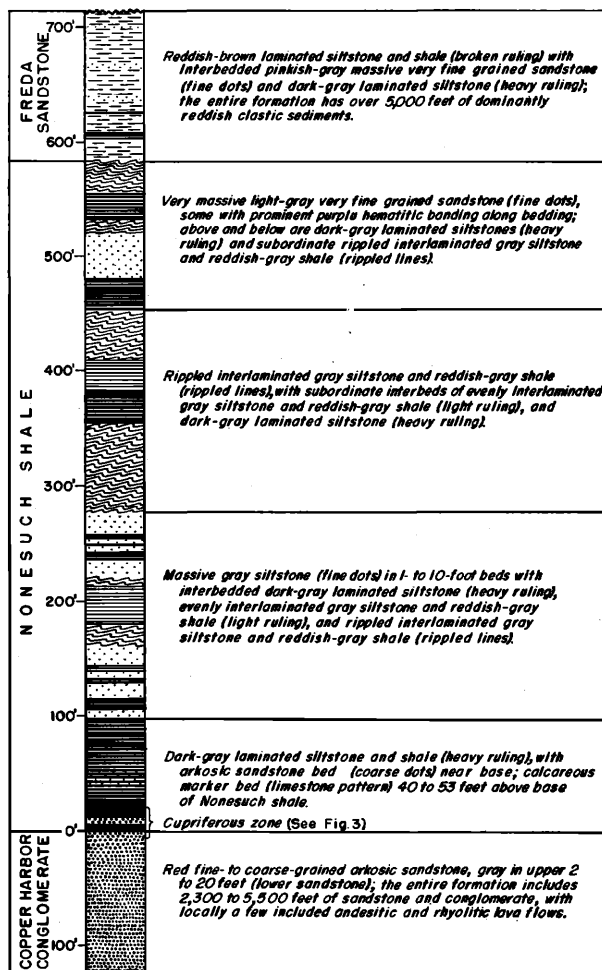


FIG. 2. Columnar section of Nonesuch shale. Section measured on Big Iron River 0.5 to 1 mile from mouth, in sec. 13, T. 51 N., R. 42 W. (locality E, Fig. 1). Bottom 20 feet of Nonesuch shale is from little Iron River (locality D, Fig. 1), and Freda sandstone is from drill hole near lake shore 0.3 mile east of mouth of Big Iron River.

Broderick, Chief Geologist of Calumet & Hecla, Inc., for much valuable information on the old White Pine mine.

Members of the Geological Survey who have provided material assistance in the work and who have contributed much by discussion of the problems involved include H. R. Cornwall, R. W. Swanson, and E. S. Davidson. Finally, the authors are most indebted to Ralph S. Cannon, Jr., of the Geological Survey, who, before the study began, suggested the possible sedimentary origin of the copper, and whose interest in the study has been a constant stimulation.

THE NONESUCH SHALE

The upper Keweenawan Nonesuch shale (5, p. 220–224) is a sequence of dominantly gray, well-bedded siltstone and shale that is distinct from the red to brown coarser grained rocks of the formations above and below. The thickness ranges from 250 feet at the Bad River north of Mellen, Wis., to more than 700 feet near Calumet, Mich. (Fig. 1), but is close to 600 feet over most of the intervening distance. The formation is separated by a sharp boundary from the underlying Copper Harbor conglomerate (7, p. 680, 690) and is more or less transitional into the overlying Freda sandstone. Figure 2 is a generalized columnar section of the formation from a point near the mouth of the Big Iron River ("E," Fig. 1).

The Nonesuch shale crops out on the north side of the Keweenawan trap range, which is upheld by the Portage Lake lava series and generally dips 15° to 45° northward or northwestward toward Lake Superior. The slightly sinuous course of its outcrop is interrupted by the structural dome of the Porcupine Mountains (Fig. 1), around which the outcrop belt makes an almost-closed loop; the beds dip rather gently (4° to 25°) away from Porcupine dome, except along the south flank, where the dips are steep and locally even overturned. The formation apparently disappears westward largely by a change of sedimentary facies. On the Bad River, just north of Mellen, Wis. (Fig. 1, inset map), the Nonesuch shale is represented by about 80 feet of gray laminated siltstone and shale beds that are interstratified with twice as much reddish sandstone and shale to form a 250-foot sequence; these gray rocks probably represent tongues that pinch out completely a little farther west. The northeastern end of the outcrop belt is at the lake shore 7 miles north of Calumet, Mich. (Fig. 1, inset map). The formation there is coarse grained and somewhat redder than in the central part of the outcrop belt and may likewise represent a marginal facies of the Nonesuch shale.

A majority of the upper Keweenawan sedimentary rocks on the south shore of Lake Superior had a southern source as shown by numerous observations of foreset beds in the Copper Harbor conglomerate and Nonesuch shale. The considerable amount of quartz and mica with minor amounts of jasper and specular hematite throughout the Nonesuch shale indicates that the streams that supplied the clastic material of the formation had their headwaters in an area of pre-Keweenawan rocks, such as those now exposed 10 miles or more south of the present outcrop. The nature of the Nonesuch rocks themselves indicates that they were probably shallow-water sediments. At least during the initial stage of deposition of this formation, the shoreline must have fluctuated back and forth across the area of the present outcrop.

The Cupriferous Zone

In the vicinity of the White Pine mine, copper is largely restricted to a distinct series of beds, about 25 feet thick, in the lowermost part of the Nonesuch shale and the uppermost part of the Copper Harbor conglomerate. For convenience of reference this group of beds will here be called the "cupriferous zone." The cupriferous zone is subdivided in present usage at White Pine

into four units, which are, in ascending order, the lower sandstone, the parting shale, the upper sandstone, and the upper shale. Figure 3 is a detailed columnar section of the cupriferous zone; the thickness of each individual unit in this section is the average for that unit (not weighted for area represented) as measured in about 75 drill cores from the White Pine drilling. The following paragraphs describe these units in ascending order.

Lower Sandstone.—In the vicinity of White Pine the sandstone at the top of the Copper Harbor conglomerate is fine- to coarse-grained without many

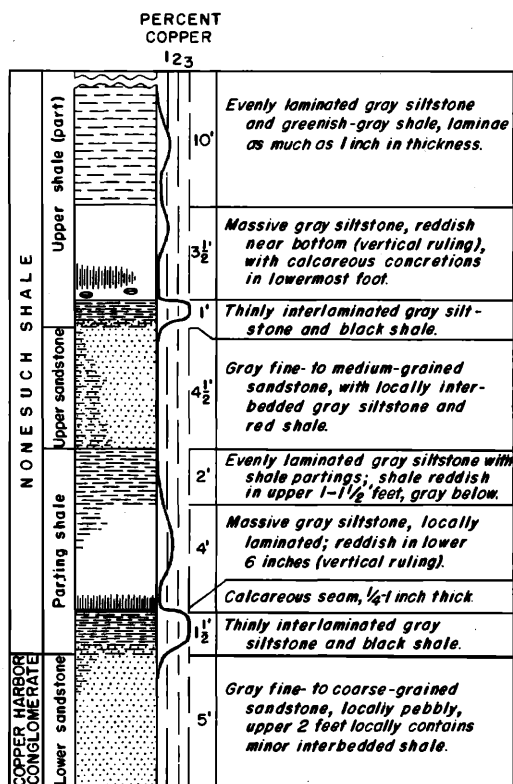


FIG. 3. Typical stratigraphic section of cupriferous zone, White Pine area, Michigan.

pebbles. The three principal granular constituents of the sandstone are fragments of mafic lava, quartz, and fragments of rhyolite, named in the most common order of decreasing abundance. The grains range from angular to subrounded. The interstices, generally 20 to 25 percent of the rock, are filled with calcite or, less commonly, with calcite and chlorite.

The uppermost 5 to 20 feet of the Copper Harbor conglomerate is generally grayish, in contrast with the prevailing reddish or brownish color of the formation as a whole; this gray unit is called the lower sandstone, and Butler,

Burbank, et al. (3, p. 172) call it the "Second lode." Comparison of thin sections representing the grayish and reddish rocks suggests that the amount of hematite is only slightly different in the two and that the major difference is a higher ratio of chlorite to calcite in the matrix of the gray rock. The brick or chocolate color of many fine-grained sandstones of the Copper Harbor conglomerate, however, is probably due largely to interstitial earthy hematite.

In the old White Pine mine, both the lower and upper sandstones locally contain considerable carbonaceous matter in the matrix (3, p. 171; 9), but it is a negligible constituent of the sandstones over most of the area drilled by Copper Range Co.

The lower sandstone locally contains copper minerals in the uppermost 1 to 7 feet and was mined at the old White Pine mine. It is almost barren over most of the area drilled by the Copper Range Co.

Parting Shale.—The parting shale, despite its name, is dominantly siltstone. Both the thickness of the unit and the sequence of beds within it, as represented in Figure 3, are remarkably uniform over all but the southeastern quarter of the 10 square miles drilled by Copper Range Co. (Fig. 1).

The boundary between the parting shale and the lower sandstone is generally marked by a transition zone 3 to 6 inches thick. This transition rock consists of fine- to medium-grained gray sandstone with numerous irregular and discontinuous thin black-shale partings.

Above the transition zone is a bed 1 to 2 feet thick composed of thinly laminated gray siltstone and black shale; most individual laminae consist of a lower siltstone portion that grades upward into a black shale parting. Individual laminae are mostly less than 0.1 inch thick, though a few may be an inch or more; many of the shale partings are only films about 0.01 inch thick. The shale at the top of most laminae is sharply set off from the siltstone of the next succeeding lamina, and it is a simple matter to distinguish top from bottom in most hand specimens of this rock. Individual laminae pinch and swell, and some, at least, are not persistent laterally for more than a few feet. The ratio of siltstone to shale in the bed is typically about 2:1. The black color of the shale partings is due to the presence of carbonaceous matter.

Next above this laminated bed is a bed of massive siltstone 3 to 4 feet thick. A fraction of an inch to 2 inches above the base of this siltstone there is a remarkably persistent calcareous layer 0.2 to 1.0 inch thick. This layer has a very sharp contact at the base, cut by what look like tiny shale-filled shrinkage cracks, and a transitional upper boundary. The 6 inches of massive siltstone above this calcareous layer generally has a faintly reddish tinge, more conspicuous in artificial light than in sunlight. The bulk of the massive siltstone above this reddish zone is gray; it contains some scattered calcareous concretions and thin discontinuous calcareous layers just above the reddish zone. In the transition upwards from this massive bed, laminations become increasingly conspicuous. Contorted bedding and angular unconformities locally indicate some contemporaneous slumping during deposition.

The next higher subdivision is commonly composed of graded laminae $\frac{1}{2}$ to 2 inches thick; dark-gray shale or mudstone at the top of each lamina grades down into gray siltstone below and has a sharp contact with the siltstone of

the next higher graded lamina. Individual laminae are persistent laterally for many tens of feet.

The next higher unit, the uppermost subdivision of the parting shale, is like the one immediately below except that the finer grained tops of individual laminae are conspicuously reddish. The coarser siltstone part of each pair may be gray, as in the unit below, or red.

The top of the parting shale locally has abundant mudcracks, and in some places this horizon is incised to depths of a foot or two by channels filled by the overlying upper sandstone.

The laminated bed at the base of the parting shale generally contains 1 to 5 percent copper as chalcocite and native metal, and the upper part of the massive siltstone bed typically contains 1 to 2 percent copper (Fig. 3).

Upper Sandstone.—The upper sandstone is very similar in appearance to the lower sandstone. It is generally a little finer grained, and pebbles are even less common than in the lower sandstone.

This bed shows more conspicuous change from place to place than do the other beds of the cupriferous zone, not only in thickness but also in lithology. In some places the upper sandstone is a single massive bed of more or less even-grained sandstone. In others it consists of several conspicuously graded beds a few inches to over a foot thick. Individual graded beds may contain material ranging in size from coarse pebbly sand to gray siltstone or red shale. Some graded beds contain nothing coarser than siltstone and are identical with layers in the uppermost parting shale beneath. The base of the upper sandstone is arbitrarily drawn at the base of the lowest sandstone layer.

The upper sandstone locally contains chalcocite or native copper, most commonly at or near the top. Like the lower sandstone, the bed is almost barren over most of the area drilled by the Copper Range Co.

Upper Shale.—The upper shale, in a broad sense, includes the next 20 to 40 feet of rock above the upper sandstone, but only the lower 7 or 8 feet is copper-bearing. The sequence of beds at the base of the upper shale is strikingly similar to the sequence in the parting shale (see Fig. 3). At the base is a unit almost identical lithologically with the thinly laminated gray siltstone and black shale at the base of the parting shale. It is generally less than a foot thick, and grades into the sandstone below through a transition zone lithologically like the one at the base of the parting shale. Above the laminated silt and shale is a massive bed 3 to 3.5 feet thick analogous to the bed in the middle of the parting shale. This massive bed is, furthermore, locally reddish in its lower part and gray above. Instead of a thin calcareous layer at its base, however, this massive bed contains one or two horizons with spheroidal calcareous concretions 1 to 3 inches in diameter in the reddish siltstone 6 to 12 inches above its base.

Above this massive bed the rock is gray siltstone with thin and very even dark-gray shale laminae. Individual laminae can be correlated on the basis of their thickness and spacing (like varves or tree rings) for distances of at least 1,000 feet. This evenly laminated unit is 5 to 15 feet thick over most of the drilled area. Where the unit is more than 6 or 8 feet thick the shale laminae in the upper part are commonly reddish. The top of the cupriferous

zone is generally less than 5 feet above the base of this evenly laminated unit.

The laminated bed at the base of the upper shale contains 1 to 5 percent copper as chalcocite or native metal, like its counterpart at the base of the parting shale. There is typically about 1 percent of copper in the upper part of the massive bed or just above, and locally the laminated rock 2 to 3 feet above the massive bed also contains 1 percent (Fig. 3).

Rocks Above the Cupriferous Zone

The lithology of the rocks above the cupriferous zone is briefly summarized in Figure 2. Most of the Nonesuch shale consists of well-bedded to laminated gray siltstones, more or less current rippled, with features like mudcracks and mud chips that suggest shallow-water deposition.

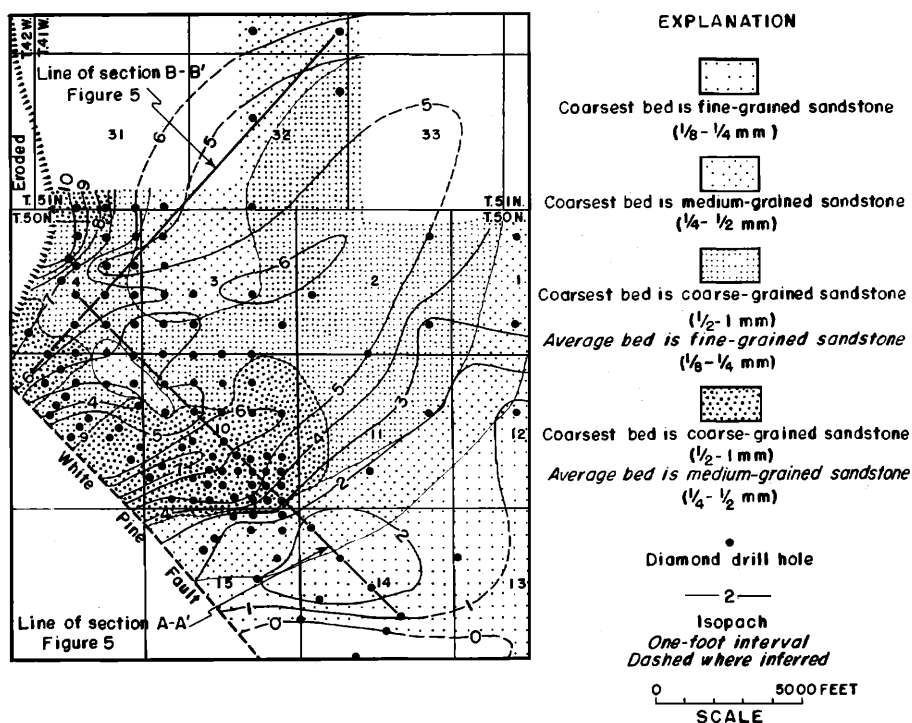


FIG. 4. Thickness and lithology of upper sandstone in cupriferous zone of the Nonesuch shale, White Pine area.

Within this series of beds one bed of unique lithology, called the marker bed, has outstanding value in regional stratigraphic correlation. It is about 10 feet thick, and its base lies 30 to 40 feet above the top of the upper sandstone. The rock of the marker bed is thinly interlaminated gray siltstone and dark-gray to black shale; it is studded with tiny white calcite blebs, both spherical and pancake-shaped, a millimeter or less in thickness. The marker bed

overlies a 3- to 4-foot bed of massive gray siltstone. This massive bed in turn overlies 1 to 2.5 feet of thinly interlaminated light-gray siltstone and dark shale, which has a distinctive striped appearance particularly in drill core. The base of this "striped" bed is taken as the top of the upper shale.

PALEOGEOGRAPHY OF ROCKS OF THE CUPRIFEROUS ZONE

The individual stratigraphic units described above are all present over a large area near the White Pine mine, and variations in their thickness and lithology furnish some evidence for the paleogeography of the area during deposition of the cupriferous zone. The sandstones are particularly useful in this respect.

The various changes of thickness and facies are illustrated in Figures 4 and 5. Figure 4 is a map showing the thickness and lithology of the upper sandstone, and Figure 5 contains two diagrammatic stratigraphic sections of the cupriferous zone, oriented at right-angles to one another, that cross the area of the White Pine copper deposit. The shorter section of Figure 5, B-B', is 3 miles long, and the longer almost 5 miles; the vertical exaggeration is 400:1.

These sections (Fig. 5) are based wholly on diamond-drill information. The lithologic data for parts of section A-A' are incomplete. For most holes, the core from the cupriferous zone was split, and the half not sent for assay is preserved for study. But all the core from the following holes in section A-A' was used for assay, and therefore cannot be examined: X-6, X-8, X-11, X-13, X-19, 8-M, 9-N, 11-P. For these holes the writers have had to depend on the original drilling record, which generally distinguishes only sandstone, shale, and mixtures of the two. The only uncertain correlation lines on the diagram, however, are those that tie the four southeasterly drill holes to the rest of section A-A'.

The datum selected for the sections of Figure 5 is an average of the position of three surfaces, namely the base of the striped bed, the top of the laminated bed at the base of the upper shale, and the top of the laminated bed at the base of the parting shale. To determine the depth of the average surface in any hole, the depths of the three surfaces were simply added together and divided by three. This average surface, if drawn on Figure 5, would be represented by a horizontal line 29 feet above the base lines of section A-A' and B-B'; all other bedding planes are drawn at their appropriate distances up or down from this average plane. If one actual surface had been used as a datum, instead of the average plane, all its own irregularities would be carried over to maps or diagrams of other surfaces that were measured up or down from it, but using an average surface reduces the magnitude of such irregularities.

Sandstones

Figure 4 shows the thickness and lithology of the upper sandstone in most of the area drilled by the Copper Range Co. This bed is thick and coarse along a line that trends northeast across the midpoint of sec. 10, T. 50 N., R. 42 W. Where this line crosses the southeast drift from the Schacht shaft

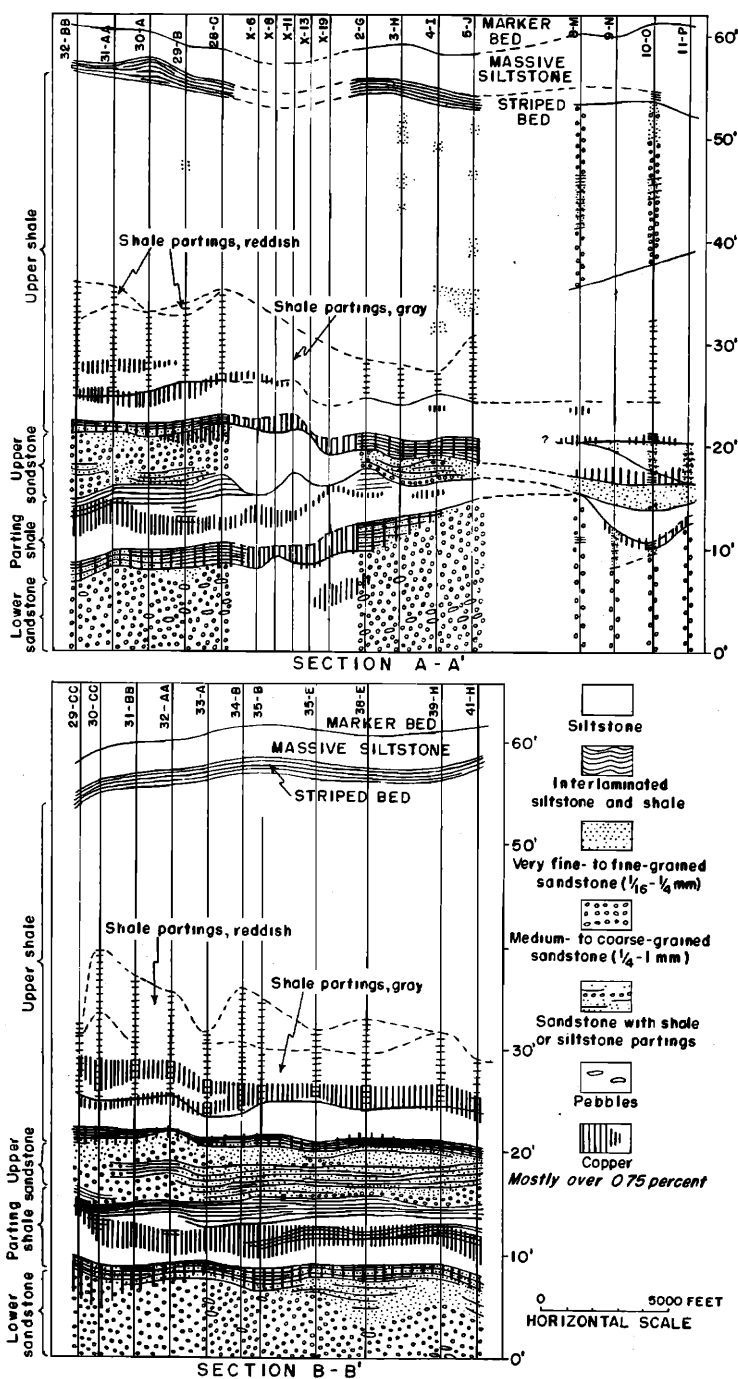


FIG. 5. Stratigraphic cross sections of White Pine area. For location of sections see Fig. 7.

(SW $\frac{1}{4}$ sec. 10), channels as much as 2 feet deep trending N. 50°–60° E. cut the upper part of the parting shale. The upper sandstone thins and becomes finer grained northeastward along this channel (Fig. 4) and also southeastward away from the channel. Both the tonguelike form of this body of sandstone and the presence of mudcracks and channels at its base suggest that the body is deltaic, deposited in shallow water, or perhaps even in part subaerial. The upper sandstone becomes still thicker to the northwest, and thus suggests that the channel through sec. 10 may be only one distributary in a system of channels feeding a large delta; another channel seems to lie just west of the drilled area.

The topography of the top of the lower sandstone probably had some such form as is shown by its profile in Figure 5. This form is not greatly changed if any of several individual beds are assumed to be horizontal. The outstanding feature of this surface is the large prominence near the eastern end of section A-A' (Fig. 5, drill holes 5-J and 8-M). There is not enough information to define the top and eastern side of this prominence, but it is interpreted as a constructional form built up along a channel in a sandstone delta analogous to the body of upper sandstone. The highest points of this deposit would be the levees containing the channel itself. The stratigraphy of drill hole 10-O suggests that the prominence separates two low-lying areas.

Thinly Laminated Beds Just Above Sandstones

The thickness and character of thinly interlaminated gray siltstone and black shale at the base of the parting and upper shales are systematically related to the tops of the sandstone beds that underlie them. In general, the siltstone-and-shale facies is thickest and most free from sandstone laminae in relatively low places near but down the slope from the relatively high places here interpreted as ancient deltaic tongues.

The laminated bed at the base of the parting shale is 1.5 to 2.5 feet thick and sand-free in drill holes 29-B, 28-C, and probably in holes X-6 to X-19 (section A-A', Fig. 2); these characteristics prevail for a mile northeast of the section, beyond which point the bed becomes sandier though not thinner. The bed is thinner and sandy to the northwest and thicker and sandy to the southeast.

The laminated bed at the base of the upper shale is thickest and relatively free from sand laminae in holes 2-G to 5-J (section A-A', Fig. 5), southeast of the prominent tongue of upper sandstone in section 10, T. 50 N., R. 42 W. (Fig. 4); the axis of this sandstone tongue goes through drill hole X-6 in section A-A', Figure 5. In general the laminated bed at the base of the upper shale is thin where the upper sandstone is thick, and thick where the sandstone is thin throughout the area. It thickens from 6 inches or less in the northwestern part of the area of Figure 4, where it is relatively sandy, to 2 feet in the southeast corner.

Massive Siltstone

The massive siltstone beds in both parting shale and upper shale also thicken gradually away from elevations. The massive bed in the parting

shale thickens westward, away from the prominence at the top of the lower sandstone. The massive bed near the base of the upper shale thickens eastward away from the thickest and coarsest part of the upper sandstone. Evidently there was a tendency for these siltstone beds to fill in depressions and subdue the relief that existed before their deposition. This result was brought about in part, at least, by slumping; contorted bedding has been observed underground in the lower of the two beds.

These massive siltstone beds represent an abrupt change of some sort from the conditions that prevailed during deposition of the underlying thinly laminated beds. Deepening of the water is suggested as a cause. Inundation of the levees or bars protecting the stagnant depressions would make the water more open and expose the bottom to more active agitation by waves. The reddish color of the lower part of both massive beds suggests that the waters were at first well aerated and oxidizing as a result of this agitation. The gray color and preservation of some laminae in the upper part of the beds, on the other hand, indicate a tendency to revert to quieter, more stagnant conditions with the passage of time; these conditions could have been brought about by still further deepening, or by rebuilding of the levees or bars protecting the depressions.

Summary

In summary, there is here a repeated sequence of beds, (1) sandstone, (2) thinly interlaminated shale and siltstone, (3) massive gray siltstone, reddish near base, and (4) evenly laminated siltstone, with gray shale partings below and red above; the repetition of this sequence represents cyclic sedimentation. The sequence of changes up to and including the massive gray siltstone can be explained as the consequence of progressive inundation of a sand plain, with deltas of sand built out beyond the main shoreline at the mouths of streams. The extent to which these deltas were subaerial is not known, but the silts and shales of the parting and upper shales are believed to have been deposited in the lagoons, ponds, or swamps between separate distributaries.

The upper sandstone, which initiates the second cycle, was probably deposited in shallower water than the massive siltstone below it, as mudcracks are locally abundant at its base. But whether the shallowing was the result of sedimentation or of a relative lowering of water level is not readily determined. There was evidently some shifting of watercourses between the first and second cycles; the lower sandstone was built up highest in the southeastern part of the area, whereas the thickest and coarsest deposit of the upper sandstone is in the northwestern part.

PERSISTENCE OF UNITS AWAY FROM WHITE PINE

Each of the main stratigraphic units found in the vicinity of White Pine has been found at other localities up to 20 miles from White Pine, east, north, and west of the Porcupine Mountains (Fig. 1). Figure 6 shows diagrammatically the units recognized at various localities in which all or part of the cupriferous zone or the overlying marker bed is exposed. The letter symbols identifying

individual stratigraphic sections are also shown on Figure 1, the index map.

The lower parts of the sections near Lone Rock (C) and on the Presque Isle River (A) cannot be reliably correlated with those east of the Porcupine Mountains (D-J). Correlations between individual sections east of the mountains, however, are relatively reliable. As shown by Figure 6, the major subdivisions of the cupriferous zone (upper and lower sandstones, upper and parting shales) can be identified at most places where exposures are adequate

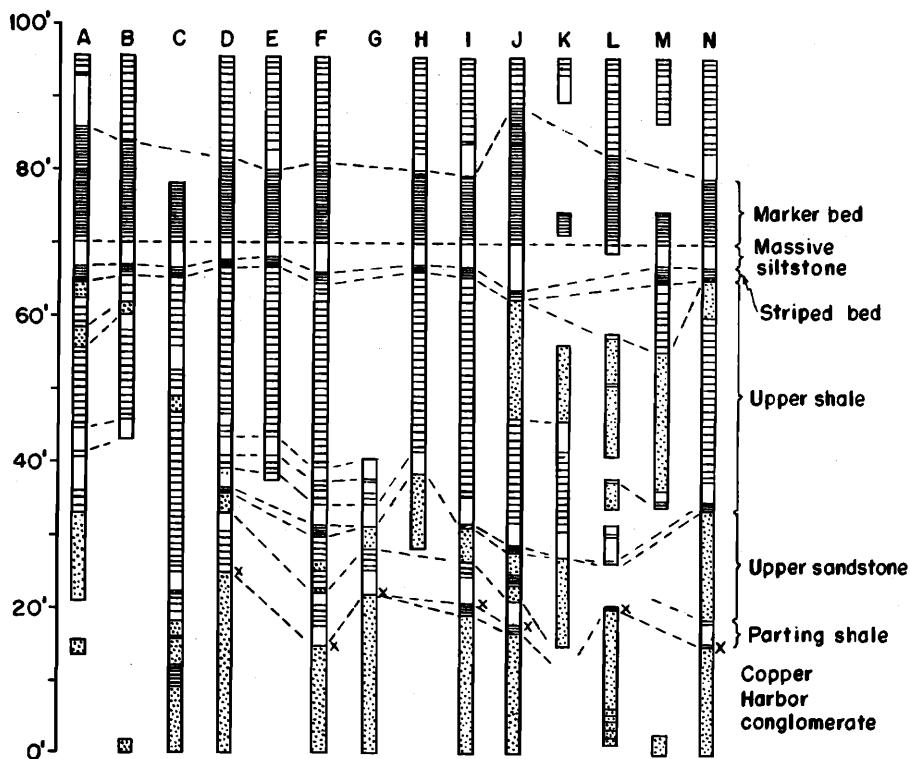


FIG. 6. Columnar sections of the lower part of the Nonesuch shale. Localities are shown on the map, Fig. 1. Patterns are as follows: stippling, sandstone; blank, massive siltstone; widely spaced lines, laminated siltstone; closely spaced lines, thinly interlaminated siltstone and shale. The presence of the calcareous layer at the base of the massive bed in the parting shale is indicated by "x." Black to red shale is also abundant 15 to 39 feet below the lowest shale shown on section C (near Lone Rock).

for 23 miles east from Nonesuch mine (H, Fig. 1) to McGuire Creek (N), and for at least 5 miles north (D) and 4 miles southeast (J) of White Pine (I). The same subdivisions were also recognized (1, p. 58-60) in drilling and underground development in the vicinity of the White Pine Extension mine (sec. 7, T. 50 N., R. 43 W.), 8 miles west of White Pine.

The thin calcareous seam at the base of the massive bed in the parting

shale (indicated in Fig. 6 by "X" in sections where it was found) was identified in practically all sections where the base of the Nonesuch shale is exposed. A calcareous layer of the same character was common in fragments on the dump (now mostly removed for use as road metal) of the White Pine Extension mine (8 miles west of White Pine), which had two long development drifts on the parting shale. This thin layer, therefore, seems to be more or less persistent for a minimum distance, from east to west, of 28 miles.

In the rocks above the cupriferous zone, the persistence of the marker bed and of the underlying massive and striped beds is shown in Figure 6. The presence of sandstone at or a little below the striped bed should be noted in the two westernmost and the five easternmost sections.

Beyond the limits of Figure 1, present exposures do not afford a basis for any bed-for-bed correlation either west of the Presque Isle River (A) or northeast of McGuire Creek (N). A thin bed with a few calcareous blebs like those of the marker bed was found 75 feet above the base of the Nonesuch shale at the mouth of the Black River, 5 miles west of the Presque Isle River. Sandstone is found interbedded with siltstone at the base of the Nonesuch shale at most other localities where exposures are fair or good. It should be noted, however, that there are no good sections of the basal Nonesuch for 36 miles northeast from McGuire Creek.

In summary, the stratigraphic subdivisions of the lowermost part of the Nonesuch shale are more or less persistent for distances of 20 miles east and west of the White Pine mine. They may continue farther northeast, but exposures are poor for the next 36 miles. The environment favorable for deposition of sediments like those at White Pine therefore existed over an area many times larger than that of the White Pine copper deposit itself.

STRUCTURE

The Nonesuch shale of Michigan and Wisconsin lies on the south flank of the Lake Superior basin and dips gently northward or northwestward towards the basin at most places. Representative attitudes in the central part of the outcrop belt are shown by dip symbols in Figure 1. Farther southwest in Wisconsin the dips are very steep to vertical. Northeast of Greenland, Mich. (Fig. 1), dips range from 40° to 55° NW. for the first 30 miles and are mostly 20° to 30° NW. still farther to the northeast.

The south flank of the Lake Superior basin is interrupted by the uplift now marked by the Porcupine Mountains (Fig. 1). The Nonesuch shale dips, in general, away from the uplift, though locally it is disrupted by faults as on the east flank of the uplift or overturned as along the south side. Between this uplift and the main Keweenaw trap range to the south are two synclinal areas underlain by the Nonesuch shale and the younger Freda sandstone. The syncline south and southeast of White Pine and Nonesuch (I and H in Fig. 1) will be referred to here as the Iron River syncline, and the other, crossed by the lower course of the Presque Isle River, will be called the Presque Isle syncline. The base of the Nonesuch shale probably attains depths of 3,000 to 5,000 feet below the surface in the Iron River syncline. Along the line of the

Presque Isle River the base of the Nonesuch is probably nowhere more than 1,200 to 1,300 feet below the surface, though the depth must increase to the northwest along the axis of the syncline.

Faults

The Nonesuch shale is much broken in places by faults. Some fairly large faults, with offsets ranging from a quarter of a mile to a mile, are shown in Figure 1. One of these, the White Pine fault described below, is the major tectonic feature of the White Pine area; it trends northwest almost bisecting T. 50 N., R. 42 W. (Fig. 1). Two faults trending east-northeast are shown offsetting the base of the Nonesuch shale between White Pine (I) and the lake shore. Another, marked by a brecciated zone almost 500 feet wide at the lake shore, is shown near Lone Rock at locality C in Figure 1.

Minor faults, too small to show on Figure 1, with displacements ranging from a foot or less to many tens of feet, are seen in stream beds where exposures are good. Most of the small minor faults have a little gouge or breccia. Secondary minerals, principally calcite, are abundant in some; quartz, and even traces of copper or chalcocite are also found. Faults with dips of less than 60° are not common, and many are nearly vertical. The larger minor faults are generally marked by a break in the continuity of outcrop, and the fault planes themselves cannot be directly observed.

Folds

Folds range in size from less than a foot to 6 or 7 miles from crest to trough. Most of the folds observed in the field and underground are bends due to drag along faults. Faults without drag are more common than those with drag, but it is not unusual to see the bedding bent up or down, or even more complexly crumpled, within a foot or two of a small fault. The disturbed zone may extend for several feet out on one or both sides of a fault with 20 or 30 feet of displacement. The beds exposed in the Big Iron River are much crumpled for several hundred feet near the two large faults shown north of White Pine in Figure 1. Even the anticline at White Pine (Fig. 7) is most simply interpreted as the result of dragging down of the beds on the northeast side of the White Pine fault.

In the curved parts of folds the rocks—whether shale, siltstone, sandstone, or conglomerate—are normally much fractured. They are cut by many tiny faults that lack notable continuity. Bending without breaking—continuous deformation—is the exception, even in the most shaly beds.

Structure Near White Pine

White Pine Fault.—The structure of the Nonesuch shale in the vicinity of White Pine is dominated by the White Pine fault (Fig. 7). This fault strikes northwest and dips steeply northeast at the old White Pine mine (3, p. 172). Its trace to the southeast, as shown in Figure 7, is based on the reasonable assumption that it underlies a mile-wide gap in the hills to the southeast of

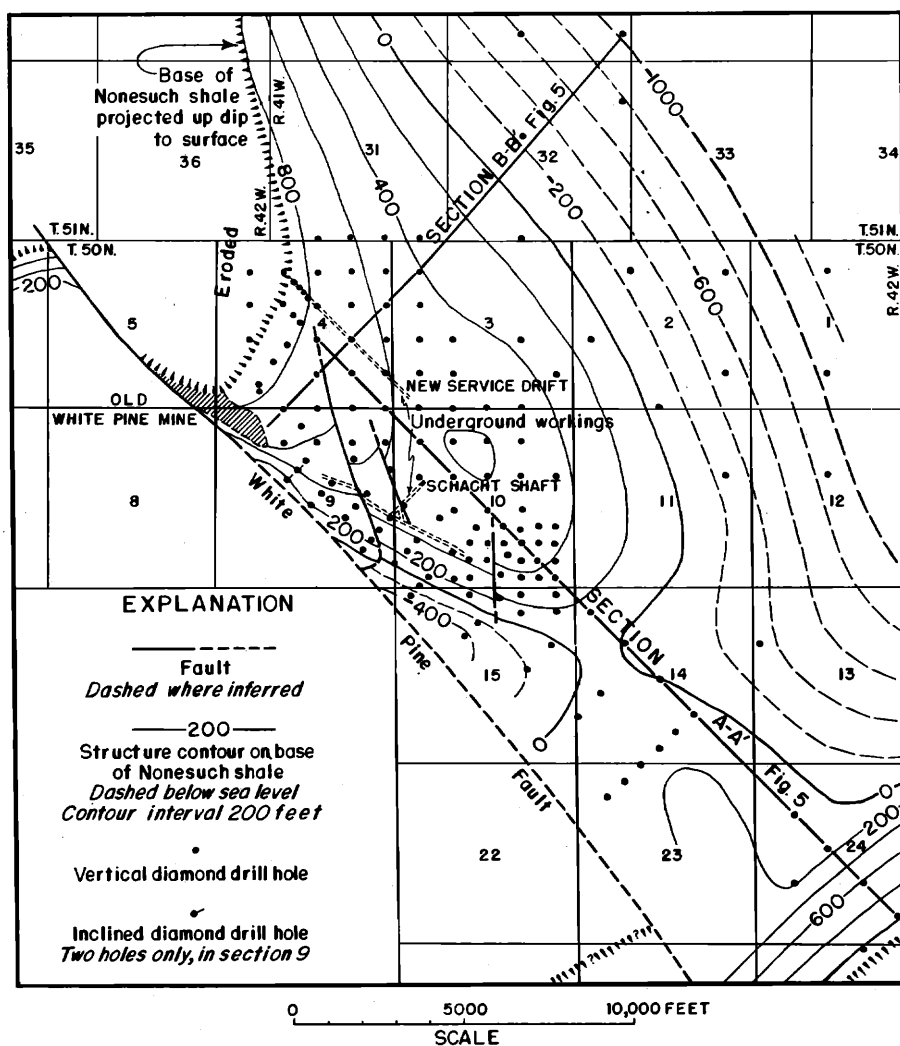


FIG. 7. Structure contour map of the White Pine area. This map shows all of the holes drilled at White Pine between 1937 and 1950. Individual holes are shown on the other maps (Figs. 4, 9, 10, 11) only if they provided data used in the construction of that particular map.

the map area. On the northeast side of the fault the position of the base of the Nonesuch shale is accurately known from drilling from one limb of the Iron River syncline to the other; the structure contours on Figure 7 depict its subsurface topography with greater or less assurance as the drilling is dense or sparse.

Immediately southwest of the White Pine fault, the position of the base of the Nonesuch shale is known at few points. It has been located by drill holes

at the old White Pine mine and northwest of that mine at the western edge of the area of Figure 7. The outcrop of the base of the formation, as shown in sec. 23, T. 50 N., R. 42 W. (Fig. 7), is based on a tenuous projection from two drill holes farther southwest, and may be in error by as much as half a mile. But even allowing for an error of this magnitude, the White Pine fault must be primarily a right-hand³ tear fault. The horizontal component of slip along the fault would amount to over a mile, and the vertical component may not be more than a few hundred feet.

Anticline.—The structure contours of Figure 7 show the prominent anticline northeast of the fault. In this fold, which is probably representative of many folds that are due to drag on faults, the curvature increases at an accelerating rate toward the fault. Along a horizontal line through the Schacht shaft, at right angles to the fault trace, the rate of change of dip is about 1° per 1,000 feet at 9,000 feet from the White Pine fault, 1° per 500 feet at 6,000 feet, 1° per 150 feet at 3,000 feet, and about 1° per 80 feet at 2,000 feet. Presumably the rate of change of dip decreases at a similar rate southwest of the fault in a synclinal fold.

The crestline of the fold shows several irregularities (Fig. 7). From a slight dome northeast of the Schacht shaft the crest descends southeastward to a prominent saddle in section 14, T. 50 N., R. 42 W. This saddle represents the intersection of the anticline with the trough of the Iron River syncline. The amount of drilling southeast of the saddle is insufficient to properly define the structure near the southeastern outcrop.

In the northeastern part of the drilled area the rocks dip northeast at a relatively uniform angle of about 10°. The fold proper should probably be regarded as ending at about 10,000 feet from the fault, where the curvature of the bedding measured in sections striking northeast becomes negligible.

Minor Faults.—The minor faults of the White Pine area have been a subject of special concern because of speculation about their effect on mining operations. They also have a bearing on the interpretation of copper distribution as recorded in drill holes. It is shown in the following that the minor faults are related to the White Pine fault, and that they, like the curvature of beds in the White Pine anticline, decrease in magnitude away from the White Pine fault. This decrease provides the basis for a test of the relationship of copper distribution to structure, discussed in a later section.

The faults exposed in the drifts at the Schacht shaft strike between N 20° E and N 50° W and have dips between 70° W and 60° E. Most strike a little west of north and dip more than 80°, either east or west. The vertical component of slip on practically all faults shows normal-fault movement and ranges in magnitude from less than a foot to about 35 feet. More than half of the faults, however, are primarily strike-slip faults. Those with right-hand slip have an average strike of N 33° W, and those with left-hand slip an average strike of N 15° W. This difference in strike suggests that the two sets may represent complementary shears so disposed that the axis of principal shortening lies in the acute angle between them, or somewhere near N 25° W.

³ As used here, "right-hand" movement means that the displacement of the block across the fault from the observer is to the right as he faces the fault.

This orientation is consistent with right-hand movement on the main White Pine fault (Fig. 7).

Maps of the old White Pine mine reveal a tendency for faults to trend more and more westerly as they are followed toward the White Pine fault; the fault in the southeast corner of section 9 (Fig. 7) has been curved to show the resulting hook-shaped pattern on the assumption that the habit persists away from the old mine. This curvature in plan also suggests right-hand movement on the White Pine fault.

The minor faults are clearly related to the White Pine fault in frequency as well as habit. The frequency of faults encountered in the mile-long drift at the Schacht shaft (Fig. 7) bears a very simple empirical relation to the magnitude of their vertical stratigraphic displacements. The average distance between faults with more than any given displacement is approximately 100 times the given displacement. For example, in a drift more or less parallel to the White Pine fault and 1,500 feet from it (the average distance of the Schacht shaft drift), faults with more than 10 feet of vertical displacement may be expected about every 1,000 feet, and faults with more than 1 foot of vertical displacement may be expected every 100 feet.

Faulting was much more intense in the old White Pine mine, which is considerably closer to the White Pine fault. Along a line 600 feet from the fault the average distance between minor faults with more than a given vertical displacement is only about 24 times the average displacement. At 500 feet the factor is about 18; at 400 feet, 14; at 300 feet, 12; and at 200 feet, 10. These empirical factors for the old mine are based primarily on the larger faults with displacements of over 5 or 10 feet as shown on an unpublished map of that mine by Dr. T. M. Broderick.

Summarizing the data on minor faults in the White Pine area, most of the faults seem to be characterized by strikewise movement. Two complementary sets of such faults may be identified, and they suggest an axis of shortening oriented about N 25° W; this is compatible with dominant right-handed strikewise movement on the White Pine fault. The tendency of minor faults to hook westward in plan as they approach the White Pine fault is likewise attributed to right-handed drag on the fault. Finally, the frequency and magnitude of minor faults decrease progressively away from the White Pine fault. The minor faults within several thousand feet of the White Pine fault, therefore, are clearly offspring in a sense of this major fault.

The progressive decrease in frequency of fractures away from the fault is paralleled by a progressive decrease, described earlier, in the rate of change of dip of the beds (i.e., a decrease in the amount of curvature). At any place the abundance and average displacement of faults are probably in fairly direct ratio to the amount of local curvature indicated by the contours of Figure 7. There are probably few faults of any consequence in the northeastern part of the area, and the greatest number should be expected on the southwestern limb of the anticline. The sharp change in dip in sec. 24 (Fig. 7) probably indicates strong faulting. The S-shaped curve in the sea-level contour in sections 11 and 14 suggests more faulting than might be expected this far from

the fault, as does the dome on the crest of the anticline just northeast of the Schacht shaft.

GEOLOGY OF THE COPPER MINERALS AND RELATED MINERALS

The principal copper minerals in the lowermost Nonesuch shale over the area drilled at White Pine are chalcocite and native copper. Bornite, covellite, and chalcopyrite are present only in infinitesimal quantities. A very small amount of native silver is generally present with the copper minerals, and pyrite was locally noted in fissures and in certain layers in and above the cupriferous zone.

Relative Abundance of Minerals

The proportions of the principal minerals show considerable variation. In mill concentrates from the old White Pine mine, sampled in 1918, 98 percent of the copper was native, and 2 percent occurred as sulfide (3, p. 106). These concentrates presumably came from the upper and lower sandstones, the ore horizons in the old mine; these horizons are almost barren over most of the area explored by the Copper Range Co. In recent pilot-mill flotation feed of parting shale from the Schacht shaft McLeod (8) established by extensive grain counts that 87 percent of the copper occurs as chalcocite and only 13 percent as native copper.

Analyses for sulfur as well as copper were made by the Copper Range Co. on samples from a number of scattered drill holes mostly located in sections 3, 4, and the northern half of 10, T. 50 N., R. 42 W. (Fig. 7). The stratigraphic units sampled are comparable in about 20 holes. These analyses indicate that the amount of sulfur, by weight, generally ranges from 18 to 23 percent of the amount of copper in the parting shale, from 20 to over 40 percent in the upper sandstone, and from 20 to 30 percent in the upper shale. In general the proportion of sulfur increases towards the east in the shale beds; the analyses of the upper sandstone do not show a clear trend. The weight of sulfur is about 25 percent of the weight of the copper in chalcocite, so these analyses indicate a high ratio of chalcocite to native copper throughout the area sampled, and sulfur in excess of that required for chalcocite in some individual samples, particularly in the upper sandstone. This excess is due in part, if not wholly, to pyrite, which can be seen megascopically in some of the cores that show excess sulfur.

In general an eastward increase in sulfur content relative to copper can be at least qualitatively recognized in the drill core itself. For example, in the upper shale along the line of section A-A' (Fig. 5), native copper was seen only in the two most northwesterly holes, and pyrite is prominent in drill holes 2-G, 4-I, and 5-J.

McLeod's (8) study of pilot mill products from parting shale at the Schacht shaft indicates that copper sulfides other than chalcocite are quantitatively negligible there. Extensive grain counts did not disclose measurable quantities in the flotation feed. In the concentrates, there was enough to determine that about 0.0014 percent of the copper occurs as bornite, 0.0010

percent as covellite, and 0.0005 percent as chalcopyrite. Traces of chalcopyrite and bornite, but not covellite, have been seen underground and in drill core; they occur as fillings in fissures.

Silver, largely if not wholly in the form of native metal, is everywhere a trace constituent of the copper-bearing rocks. The top of the lower sandstone was mined for native silver a little north of White Pine at the localities marked D, E, F, and G in Figure 1, and elsewhere in their vicinity, during a short-lived boom from 1872 to 1875, described by Rominger (11, p. 153-159), Whittlesey (15), and Jamison (6, p. 183-193).

The Copper Range Co. made assays for silver on the same samples that were analyzed for sulfur. The analyses from 22 drill holes show that the ratio of silver to copper is generally highest in the parting shale (silver equals 0.08 percent of copper), intermediate in the upper shale (0.03 percent), and lowest in the upper sandstone (0.02 percent). Holes with high and low silver content are interspersed irregularly over the area sampled, and there seems to be no systematic areal variation in silver content or the ratio of silver to copper.

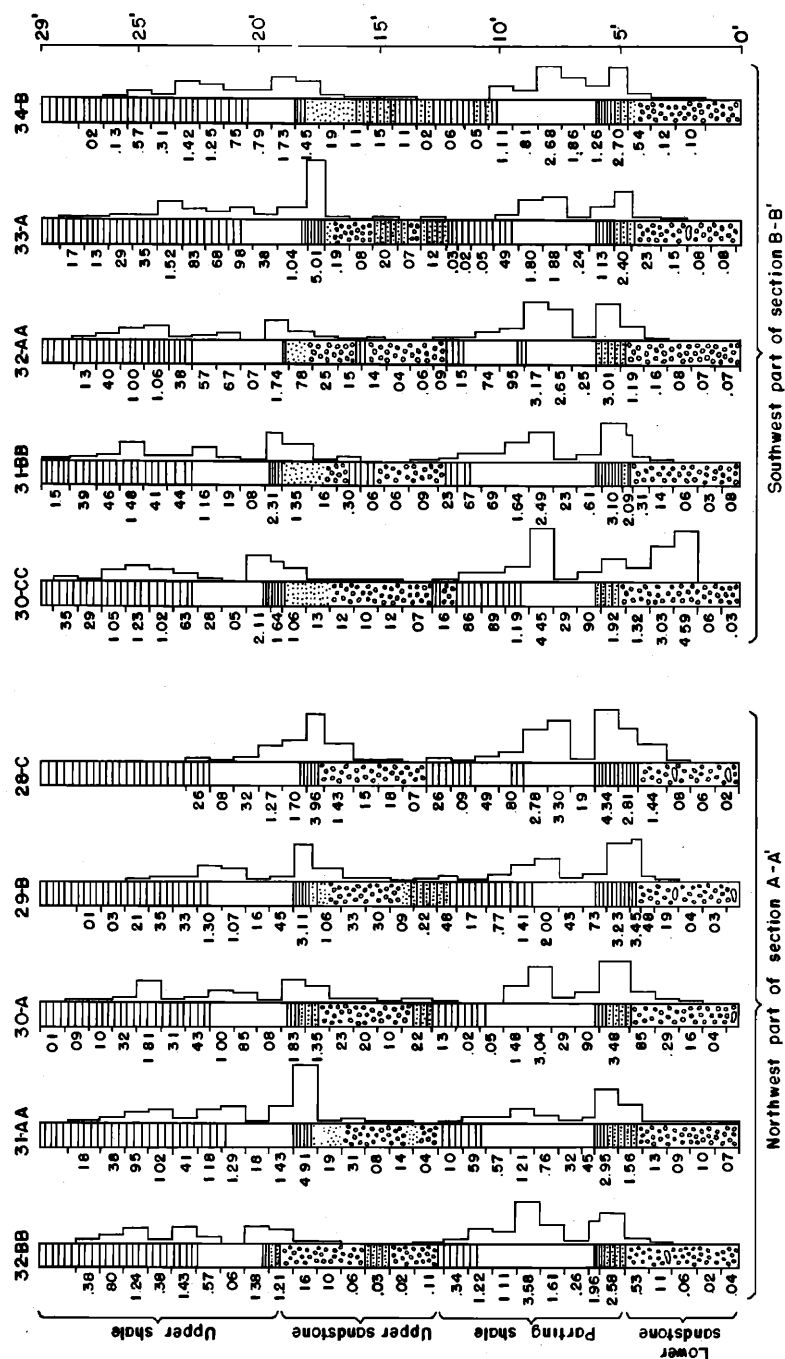
Pyrite was noted in a number of drill cores from White Pine. It occurs particularly in the laminated siltstone and shale 6 to 9 feet above the upper sandstone in the eastern part of the drilled area, where this stratigraphic zone is not copper-bearing. Traces are also visible megascopically in many cores from the striped bed, still higher stratigraphically. The pyrite has the form of tiny cubes, mostly 0.02 to 0.07 mm. across. These cubes occur as disseminated grains, or in clusters a millimeter or two across; pyrite may make up from 10 to 75 percent of the material in any given cluster (the remainder is siltstone or shale), depending on how closely spaced the individual cubes happen to be.

Stratigraphy of Copper Distribution

The areal distribution of copper in the various beds of the parting and upper shales can be roughly correlated with variations in the lithology and thickness of the beds. In general the lines of equal copper content are independent of structure; they trend northeast, more or less parallel to isopachs and facies boundaries. Copper in the upper and lower sandstones, on the other hand, is clearly most abundant near the White Pine fault.

Chalcocite and native copper occur principally as fine disseminations in shale, siltstone, and sandstone, and to a lesser extent as thin seams on joints or bedding planes and as vein minerals on faults. Chalcocite, in addition, forms small nodular masses up to several millimeters in diameter occurring individually or in groups in bedding planes; the nodules grouped in bedding planes resemble beads strung loosely together when viewed in cross section.

The abundance of these copper minerals varies systematically and strikingly from bed to bed. Figure 8 shows the assay data for the cupriferous zone in 10 holes from the western ends of sections A-A' and B-B' (Fig. 5). The sampling was done by others before the writers studied the lithology of the core, and individual samples may actually represent the rock a few inches higher or lower than the diagram shows; but the amount of such error is probably less than 3 inches at most places. As the figure shows, the stratigraphic



localization of copper ⁴ is very consistent. Virtually all the copper is found in five stratigraphic subdivisions of the cupriferous zone. These five copper-bearing subzones include two in the parting shale and three above; drill holes 32-BB and 31-BB best typify this distribution. Counting up from the bottom, the first and third subzones coincide with the laminated beds at the base of the parting and upper shales, respectively. The second subzone is near the top of the massive siltstone bed in the parting shale, and the fourth typically occurs in and just above the top of the massive siltstone bed in the upper shale. The fifth subzone, absent in some places (Hole 28-C) and merged with the fourth in others (Hole 34-B), is 6 to 8 feet above the base of the upper shale.

The assay data can be combined or represented in various ways to show the stratigraphic and areal distribution of copper for the drilled area as a whole. The grade curve in Figure 3 shows the typical stratigraphic distribution of copper in the various beds of the cupriferous zone. The grade of each stratigraphic interval is an average computed from 60 to 80 drill holes, but because this figure is intended to typify rather than summarize, this average is not a true average for the drilled area as a whole. In the first place the grade from each drill hole has not been weighted for the area represented by the hole, so undue weight is given the area of most drilling, to the west. And second, some of the subzones are absent in part of the area (see Fig. 5), and the average grade has not been diluted to take the very low copper content of these areas into account.

Figure 5 shows the stratigraphic range and lateral persistence of the individual copper-bearing subzones in two sections that cross the entire drilled area at right angles to one another. The actual assay data for five holes at the western end of each of these sections are shown in Figure 8.

Areal variation in the copper content of the rocks may be represented in at least two ways. Grade maps of a bed show the concentration of copper in that bed. A second kind of map shows the amount of copper in the bed, which is a function of both the grade and the thickness.

Figure 9 shows areal variations in the grade of the laminated beds at the base of the parting and upper shales, respectively. Most of the grades on which the contours are based, however, cannot be taken directly from the assay data, because sample boundaries do not coincide with the top, in particular, of the laminated beds in a majority of the drill holes. It is apparent from the histograms of Figure 8 that the beds overlying and underlying the laminated beds are typically very lean. In fact, the abrupt drop in grade at the top of these laminated beds is one of the most persistent and uniform characteristics of assay logs of channel samples and drill core in the White Pine area. Therefore samples that overlap the boundaries of the relatively rich laminated beds will have an intermediate grade that actually represents neither the richer nor the leaner bed. Including these overlapping samples would inevitably lower the apparent grade of the laminated beds in a majority of the drill holes, but leaving these samples out of consideration would greatly reduce the amount of information available. For the purpose of constructing a grade map, there-

⁴ "Copper," unless prefixed by "native," includes copper in chalcocite as well as native copper.

fore, it has been assumed that each overlapping sample can be split into a rich fraction representing the laminated bed, and a lean fraction representing the other. The allocation of copper to each fraction is based on an assumed value for the grade of the lean fraction. These assumed values, which are averages of the grades of the overlying and underlying beds in nonoverlapping samples, are as follows for the various beds: (1) sandstone just below parting shale, 0.5 percent copper northeast of crest of White Pine anticline (Fig. 7), 0.7 percent copper southwest of crest; (2) first foot above laminated bed at base of parting shale, 0.35 percent copper; (3) sandstone just below upper shale, 0.57 percent copper; and (4) first foot above laminated bed at base of upper shale, 0.31 percent copper. The estimated grade of the laminated bed at the base of the parting shale in drill hole 34-B (Fig. 8) can be estimated as an example of the computation. The 1-foot sample with a copper content of 1.26 percent copper is only half from the laminated bed. If the upper half contains an assumed 0.35 percent copper, the lower half contains 2.17 percent. The next lower sample, 9 inches thick, contains 2.70 percent copper. Combining these two samples gives 15 inches with an average grade of 2.49 percent copper, the grade used in preparing Figure 9B.

The three subzones in and above the massive siltstone beds of the parting and upper shales, respectively, do not have simple boundaries that coincide with lithologic contacts, and are therefore more difficult to treat in terms of grade. A given subzone may be patently richer than others in some drill holes, as can be seen from the sample data of Figure 8, but without lithologic boundaries as a guide, there is no really satisfactory basis for determining the distribution of copper in samples that may include both rich and lean material. Most of the difference between the assays of the upper copper-bearing subzone of the parting shale in drill holes 30-CC and 32-AA, for example, could be due either to the location of the sample boundaries or to actual differences in grade and grade distribution. Because of this and related difficulties, no attempt is made here to represent the grade of the higher copper-bearing subzones of the parting and upper shales.

Figure 10 shows areal variations in the amount of copper in each subzone. Data for the two uppermost subzones are combined in a single map (Fig. 10D). The amount of copper is expressed in terms of foot-percent, which is obtained by multiplying the thickness of a unit by its average grade over this thickness. Adding a foot or two of lean material beyond the limits of a subzone makes very little difference in the figure for foot-percent, so the choice of a limit is not very important. Foot-percent was chosen for convenience of calculation from the available data, because the grades of 1-foot samples are simply added to get foot-percent. But the unit can be readily converted to other measures; 1 foot-percent is equal to 1.67 pounds per square foot or 8.1 kilograms per square meter, assuming a density of 12 cubic feet per ton.

In summary, the characteristics of copper distribution are here recorded in five figures. Figure 8 shows specimens of the sample data; both this figure and the grade curve of Figure 3 show the stratigraphic distribution of copper typical of the western part of the drilled area. Figure 5 shows the lateral persistence of the copper-bearing subzones as stratigraphic units. Figure 9 re-

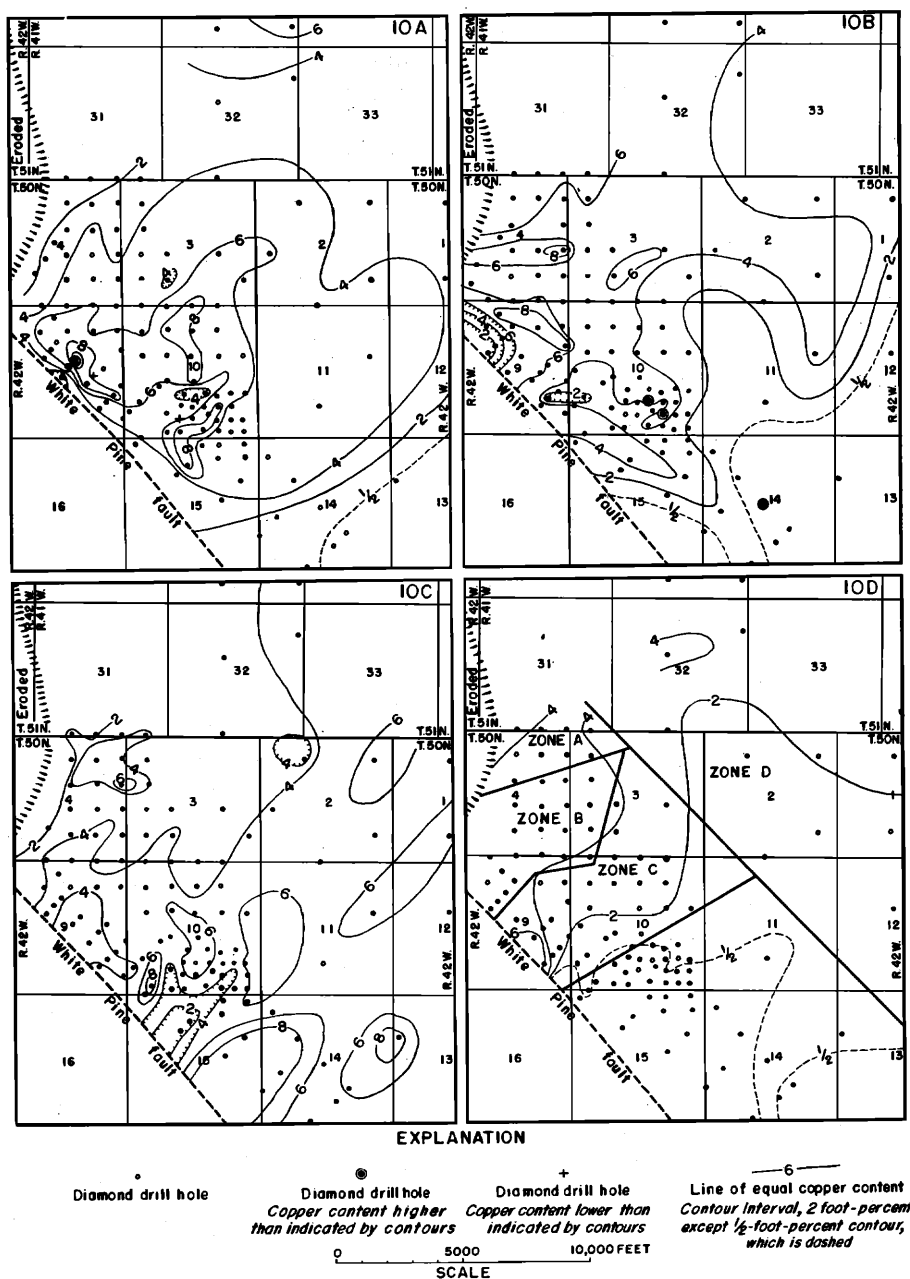


FIG. 10. Amount of copper in subdivisions of the cupriferous zone: A. Thinly laminated unit at the base of the parting shale, including copper in lean foot immediately above; B. Massive siltstone unit in the middle of the parting shale, including all copper in parting shale not shown in Fig. 10-A; C. Thinly laminated unit at the base of the upper shale, including copper in lean foot immediately above; D. Massive siltstone and overlying evenly laminated siltstone and shale of the upper shale. Zones A, B, C, and D referred to in text.

cords the thickness and grade of the laminated beds at the base of the parting and upper shales. Figure 10 shows the amount of copper in each subzone, without regard to whether this amount of copper is contained primarily in one foot or several. These five diagrams form the basis for the following descriptions of copper in the individual subzones.

Laminated Bed, Base of Parting Shale.—The bed of thinly interlaminated gray siltstone and black shale at the base of the parting shale is copper bearing wherever it is present near White Pine. Chalcocite, with some native copper, is finely disseminated all through it. Some layers have more than others, and there is a strong tendency for the greater part of the chalcocite to occur in the silt laminae. The chalcocite is essentially interstitial to the lithic minerals, and of comparable grain size (minute specks up to 0.04 or 0.05 mm). A large proportion of the chalcocite is in grains 0.01 to 0.03 mm in diameter. Individual copper-rich laminae, many of them paper thin, may be consistently and rather uniformly rich for at least the length of a thin or polished section, and the richness may be persistent for much greater distances. Others fade out laterally along the bedding.

Locally in this and the higher copper-rich subzones, some of the chalcocite occurs in small nodular clots. These clots are typically as much as 4 or 5 mm across, their centers are pure chalcocite, and the amount of impurity increases outward across their gradational boundaries. The laminations of the enclosing rock are bent up and down, respectively, above and below these nodules, either as the result of growth of the nodule, or because of differential compaction of the rock, or both. Though the original mineralogy of these nodules cannot be proved, there is no question that they formed while the rocks were still relatively unconsolidated.

A number of drill cores of this bed show one or more layers of pure chalcocite ranging from paper-thin partings to laminae 1 or 2 mm thick. Some of these chalcocite partings are probably akin to the nodules, and others are clearly postlithification bedding veinlets. Cross-cutting veinlets of similar width were locally seen cutting the thinly laminated bed in drill core, and they are relatively common, particularly near faults, in the rocks exposed in the Schacht shaft workings.

The base of the thinly laminated bed grades into the underlying sandstone. Copper is typically very abundant in transition rock that has a relatively large percentage of black-shale partings separating the thin sandy layers, but very little copper is found in the underlying rock with few such shale partings. By the same token, where the laminated bed becomes excessively sandy (Fig. 9A) the grade (Fig. 9B) and total copper content (Fig. 10A) decrease notably.

Figure 9B shows the grade of the laminated bed in percent copper, and Figure 10A shows the amount of copper in the bed in foot percent. These diagrams may be compared with Figure 9A, which shows thickness and lithology. The bed is 1.5 to 2.5 feet thick and relatively free from laminae of sand-size material ($\frac{1}{16}$ mm to 2 mm) in a lobate area trending northeast across the center of the map area. The bed is also richest in this area (Fig. 9B). Because the bed gradually thickens eastward, the total amount of copper (Fig. 10A) is highest a little east of the richest portion. And though the grade is

typically low in the exceptionally sandy material of the easternmost part of the area (The bed there is really sandstone with a few shaly partings.), the total amount of copper is moderately large because of the greater thickness of the bed.

This laminated bed at the base of the parting shale shows better than any other bed a close correlation between copper content and the depositional environment of the bed itself. The sandstone "high" that interrupts the continuity of the bed to the southeast (see drill hole 5-J, Fig. 5) trends northeast (Fig. 9A). The area of richest and most abundant copper lies down the slope northwest of this "high," and has its longest dimension parallel to the "high." The northeastward trend of the rich area, therefore, is parallel to paleogeographic features that existed at the beginning of Nonesuch time. It is at right angles to the White Pine fault and the crest of its associated anticline, instead of parallel to them as might be expected if structural control determined distribution of copper. And richness is almost inversely proportional to sandiness,

TABLE 1
COPPER IN MASSIVE BED, MIDDLE OF PARTING SHALE

Distance base of massive bed to first reddish layer above massive bed (feet)	Number of holes	Grade (percent copper) for various intervals (feet) above base massive bed					Total copper (ft-percent)
		0-1	1-2	2-3	3-4	4-5	
5½-6	22	0.44	1.29	2.28	1.33	0.69	6.03 ^b
4½-5	30	0.65	1.49	1.68	0.95	0.56	5.33
3½-4	10	0.38	0.82	1.75	0.77		3.72
2½-3	7	0.27	0.59	1.07			1.93

a condition that should hardly be characteristic if primary permeability of the bed itself were a dominant factor.

Massive Bed, Middle Parting Shale.—The gray upper part of the massive bed in the middle of the parting shale is typically copper bearing. Chalcocite with subordinate native copper is finely disseminated through the siltstone in grains comparable in size with the detrital grains, principally quartz, that make up the rock. Some of the grains of chalcocite look interstitial, particularly where groups of more or less contiguous grains form a skeletal clot within the rock. Other grains of chalcocite, on the other hand, look very little different in shape and disposition from the detrital grains of quartz and other materials.

Nodular masses of relatively pure chalcocite similar or identical to those in the laminated bed below are found also in this bed. Most of the rock is massive, and divergence of layers around the nodules was not observed.

The typical stratigraphic and geographic distribution of copper within this subzone is shown quantitatively by the assay data. Table 1 is a tabulation of the average grade, in percent of copper, of each foot above the base of the massive bed. The average grade and total copper content vary with the thickness

^b About 6.5 foot-percent if interval 5-6 feet is included.

of the massive bed plus the gray laminated shale and siltstone above it; the data of the table are arranged in order of decreasing values for this stratigraphic thickness. The rock with reddish laminae that overlies this subzone typically contains less than 0.2 to 0.3 per cent copper.

The table shows that rock from 2 to 3 feet above the base contains the most copper (30 to 55 percent of the copper in the bed), regardless of the distance to the reddish rock above, and that the bottom foot, a reddish siltstone, has very little.

Averages such as those in Table 1 could conceal considerable variation in the actual position of the peak grade from place to place, but actually the variation is not great. The average position of the peak grade in 61 holes is 2.4 feet above the base, and the standard deviation is only 0.5 foot. Both the distribution of average grades (Table 1) and the relative constancy of the position of the peak grade bespeak a very uniform stratigraphic distribution of the copper.

As can be seen in Figures 5 and 8, the position of this copper-bearing subzone is about the same whether the base of the lowest overlying laminated gray siltstone and dark gray shale is 2 or 5 feet above the base of the massive siltstone bed. This suggests that deposition of copper is independent of the degree of lamination of the rock.

The areal distribution of copper in the middle of the parting shale is shown in Figure 10B. The thickness of massive gray siltstone and the overlying gray laminated siltstone and shale is greatest (Fig. 5) and the copper in this bed most abundant in the westernmost part of the area. The water is presumed to have been deeper in this vicinity than in other parts of the area during the early part, at least, of the deposition of the massive bed (Fig. 5). Where the massive bed becomes entirely reddish (e.g., Hole 5-J, Fig. 5) it contains no copper.

Laminated Bed, Base of Upper Shale.—The thinly laminated bed at the base of the upper shale is similar in copper content, as well as lithology, to the bed at the base of the parting shale. Chalcocite and native copper have essentially the same mode of occurrence in both beds. According to the ratio of the two minerals, as computed from analyses for sulfur, most of the native copper in the upper shale occurs in the westernmost mile or less of the area; megascopically visible native copper is far less abundant than chalcocite even there.

Figure 9D shows the grade of the laminated bed at the base of the upper shale in percent copper, and Figure 10C shows the amount of copper in the bed in foot-percent. These diagrams may be compared with Figure 9C, which shows the thickness and lithology of the bed, and with Figure 4, which shows the thickness and lithology of the underlying upper sandstone. There is no such striking correlation in detail between copper distribution and sedimentary facies as was noted in the laminated bed at the base of the parting shale, but the following correlations may be noted. The bed is very thin but rich in a narrow belt just east of the area in which it consists predominantly of sandstone laminae (Fig. 9C). Another narrow belt of richer rock seems to follow the eastern margin of the area of subordinate sandstone

laminae in secs. 3 and 10, T. 50 N., R. 42 W. The most extensive areas of rich rock, however, are in the easternmost part of the area; because the bed is also thickest in this vicinity (Fig. 9C), the total amount of copper (Fig. 10C) is much greater than that which characterizes the western half of the area. Most of this eastern area lies well down the southeastern slope of an upper sandstone delta tongue (Figs. 4 and 5).

Here, as in the laminated bed at the base of the parting shale, therefore, the most abundant copper seems to occur in a place that received a thick deposit of laminated siltstone and shale, down the slope from a "high" in the underlying sandstone. It is interesting to note (Fig. 5) that outpouring of the upper sandstone seems to have completely changed the position of high and low places, and that the areas of high copper content go with the low places in the beds above both sandstones.

Copper in and above Massive Bed, Upper Shale.—Copper is found from 2 to 8 feet above the base of the upper shale. It occurs primarily as disseminated chalcocite and as chalcocite nodules, as in the lower copper-bearing subzones. Over a limited part of the White Pine area (Zone B in Fig. 10D),

TABLE 2
COPPER IN MASSIVE AND HIGHER BEDS, UPPER SHALE

Zone ^a	No. of holes	Grade (percent copper) for various intervals (feet) above base of the massive bed						
		0-1	1-2	2-3	3-4	4-5	5-6	6-7
A	9	0.29	0.29	0.95	1.11	1.04	0.75	0.49
B	19	0.19	0.34	0.92	0.79	0.61	1.06	0.72
C	21	0.28	0.33	0.92	0.91	0.41	0.32	0.10
D	12	0.12	0.23	0.80	0.61	0.46	0.26	0.07

there are two distinct copper-rich layers, both recognizable in the assays of several drill holes in Figure 8 and in the grade curves of Figure 3; they are generally 2 to 4 and 5 to 7 feet, respectively, above the base of the massive siltstone bed. The lower concentration occurs at the top of the massive siltstone bed and extends a little higher into the overlying interlaminated gray siltstone and shale; the upper concentration occurs entirely in this interlaminated siltstone and shale.

In Zone A (Fig. 10D), northwest of Zone B, these two copper concentrations seem to merge into a single subzone (drill hole 34-B, Fig. 8). The uppermost copper-bearing subzone dies out southeast of Zone B (Fig. 10D, and section B-B', Fig. 5) and is absent in Zone C. In several drill holes in Zone C and in the Schacht shaft, however, small amounts of pyrite occur in the same stratigraphic interval as the uppermost subzone; this pyrite occurs as small cubes, 0.02 to 0.07 mm across, disseminated or in clusters 1 to 2 mm across. The lower of the two subzones more or less disappears southwest of Zone C, and here, similarly, small amounts of pyrite were observed in some

^a Zones labeled in Figure 10D.

drill holes in the same stratigraphic interval as the lower subzone. Most of the drill holes in Zone D (Fig. 10D) show a copper distribution like that of Zone C or A, but a few are like Zone B.

The assay data on these uppermost copper concentrations are summarized in Table 2. The data are segregated according to the various zones of Figure 10D. The stratigraphic characteristics of the copper distribution in the various zones as well as the limited vertical range of the individual copper concentrations are apparent from these figures.

The gradual eastward disappearance of copper in the two uppermost subzones does not appear to correlate with any recognizable change in the character of the enclosing beds. On the other hand, lines of equal total copper content (Fig. 10D) trend roughly northeast, like the axis of the upper sandstone delta (Fig. 4) and the isopachs of other beds (Figs. 9A, 9C); this at least suggests a closer dependence of copper content on the environment of deposition of the enclosing rocks than on the folds and faults of the area, which trend northwest.

Copper in Sandstone.—Chalcocite and native copper occur in the lower and upper sandstones, particularly southwest of the crest of the White Pine anticline. Native copper was dominant in the old White Pine mine, and chalcocite is more abundant over the area drilled by the Copper Range Co. In the drilled area, individual chalcocite grains or groups of grains are cleanly molded around and between the detrital grains; it is clear that the chalcocite has either filled interstitial voids or has replaced some sort of matrix, either another cementing mineral like calcite (which is abundant in these rocks) or fine-grained detrital material.

The general distribution of copper in the sandstone beds, unlike that of copper in the parting shale, yields a pattern that would be expected if the copper was deposited from solutions moving up the dip from the vicinity of the White Pine fault. Figure 11 shows the amount of copper in the upper 7 feet of the lower sandstone (Fig. 11A) and in the upper sandstone (Fig. 11B). The sample that contains the uppermost few inches of a sandstone bed is not reckoned in the total for that bed, in order to make sure that no copper from the rich overlying shale bed is included; the sample containing 1.44 percent copper at the top of the lower sandstone in drill hole 28-C (Fig. 8), for example, is not included. Drill hole 30-CC (Fig. 8) provides a good example of high copper content that is entirely and unequivocally in the lower sandstone.

Most of the holes with abundant copper in sandstone are near the White Pine fault, and all but one are southwest of the crest of the anticline. The native copper mined at the old White Pine mine (Fig. 11) came from the sandstones. Copper might be more abundant on this southwestern limb of the anticline because solutions rising along the sandstone beds from the fault would not tend to move beyond the crest and down the opposite limb; or it might be more abundant because this limb is more highly faulted and bent. Though more closely spaced information on copper and faults would be very desirable, Figure 11 shows no striking concentration of copper near the three known minor faults that are large enough to be detected from the

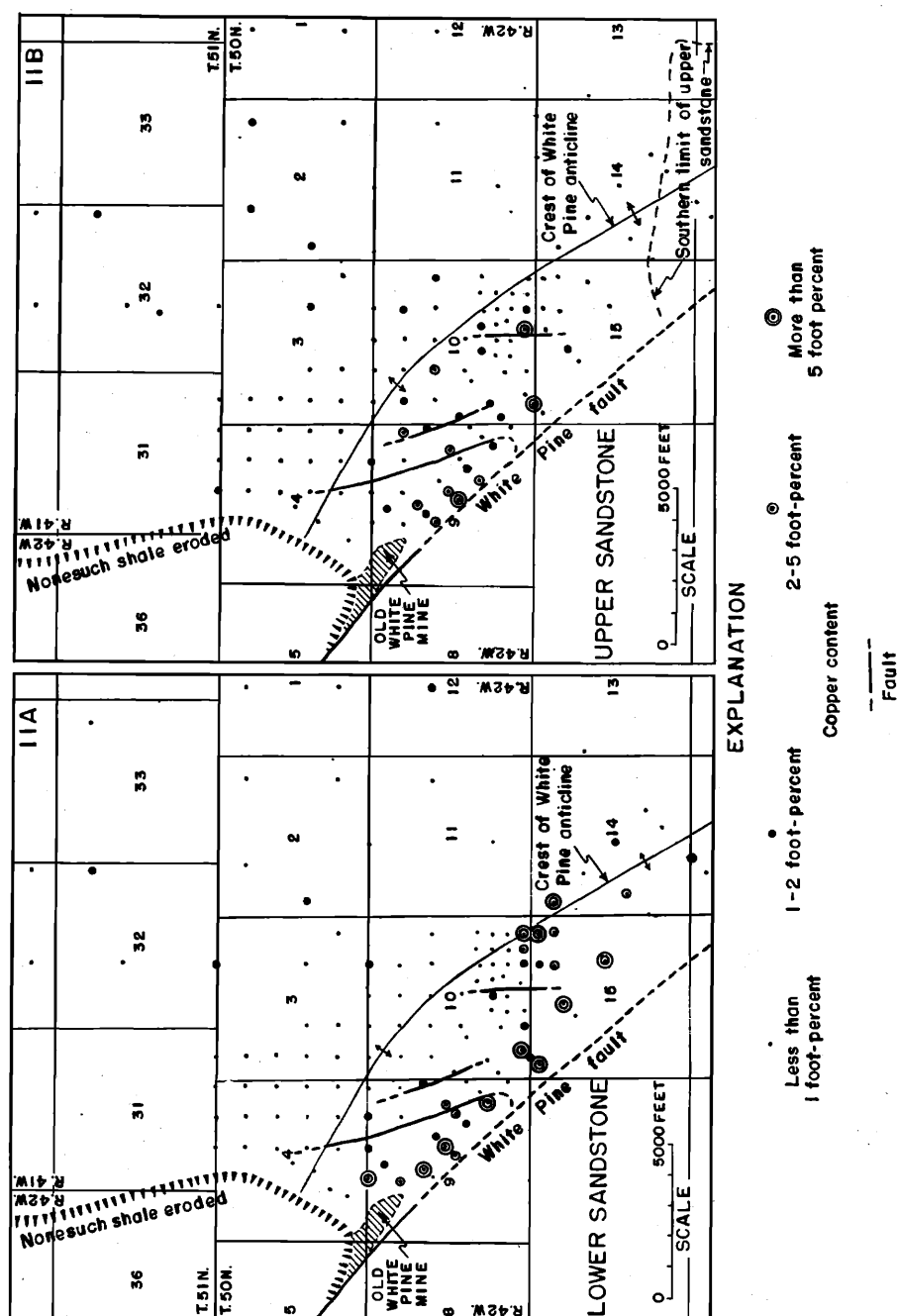


FIG. 11. Copper content of lower and upper sandstones.

drilling. One may tentatively conclude, therefore, that the principal channel-ways for copper migration were not the minor faults but the sandstone beds themselves. The apparent up-dip decrease in abundance of copper on the southwestern limb of the anticline may simply reflect increasing distance from the White Pine fault as a source of copper-bearing waters; the crest of the White Pine anticline would set a northeastern limit to the up-dip migration of these waters. There is nothing known at present to indicate whether the copper itself has a deep hydrothermal source or whether it was leached from shale beds farther down the dip or southwest of the White Pine fault.

The erratic distribution of copper in sandstone is in marked contrast to the more even distribution in the parting and upper shales. The distribution in sandstone is not amenable to contouring at the scale of Figure 11, whereas the contours of copper distribution in the shale beds (Fig. 10) show relatively few sharp peaks and hollows. If the copper in both sandstones and shales was deposited from the same general hydrothermal solution, and most of the actual flow took place in the more permeable sandstones, it is difficult to understand (1) why the patterns of copper distribution are not more similar in sandstone and shale and (2) why the patterns of distribution are not at least as regular in the sandstone as in the shale.

In summary, the distribution of copper in the sandstone beds suggests deposition from solutions moving up the dip from the vicinity of the White Pine fault; on reaching the crest of the anticline, such solutions would tend to migrate upward to the northwest along the crestline, and no appreciable amount of copper would be found northeast of the crestline. Conversely, the small amount of copper in sandstone northeast of the crestline suggests that there is no important source of solutions comparable to the White Pine fault zone down the dip to the northeast.

Relation of Copper to Fissures and Faults

Copper in Fissures and Joints.—Chalcocite and locally chalcopyrite occur as constituents of mineralized fissures, and films of chalcocite and native copper are common on joints in the copper-bearing beds. Chalcopyrite, like pyrite, tends to occur in drusy cavities in fault breccia, whereas chalcocite generally occurs in small masses or veinlets in more solidly filled calcite veins. John Rand, Geologist of the White Pine Copper Co., recovered a few small orthorhombic crystals of chalcocite by dissolving a quantity of vein calcite in acid.⁷

Films of native copper and chalcocite on joints are rather common in the parting shale in the drifts from the Schacht shaft. In the aggregate they probably contain only a small fraction of all the copper in the rock, and neither channel samples from the mine (which must include many such films) or drill-hole samples suggest that copper content of these veinlets disturbs the essentially stratigraphic distribution of the copper; the joints tend to be filled mainly where they cut the copper-bearing beds. A large number of specially chosen samples would be needed, however, to establish whether these veinlets

⁷ J. Rand, oral communication.

are lateral secretions or whether they really represent copper introduced from some distance away.

Within a few feet of some of the larger faults that cross the Schacht shaft drifts, there are numerous thin veinlets of chalcocite in the top 2 or 3 feet of the lower sandstone. Not enough channel samples from the Schacht shaft include the uppermost foot or two of the lower sandstone, however, to estimate the amount—if any—of enrichment near faults that may be due to such veinlets.

Relation of Copper in Shale to Faults and Fissures.—The amount of copper in the parting shale in the Schacht shaft drifts is independent of proximity to faults or the amount of faulting. More than 250 channel samples

TABLE 3
COPPER IN PARTING SHALE VERSUS DISTANCE FROM FAULTS,
SCHACHT SHAFT EXPLORATION

Distance from faults (feet)	Amount of copper (foot-percent) (Number of channels in parenthesis)				Standard deviation from average for exploration
	Northwest part	Middle part	Southeast part	Average for exploration	
Laminated bed					
0-2	(6) 5.4	(31) 6.5	(15) 6.8	(52) 6.5	2.2
2-4	(8) 7.5	(22) 6.4	(6) 6.5	(36) 6.6	2.4
4-8	(5) 8.1	(32) 6.5	(11) 7.8	(48) 7.0	2.4
8-16	(11) 6.2	(33) 6.0	(7) 7.0	(51) 6.2	2.4
16-32	(10) 6.9	(26) 6.5	(11) 7.6	(47) 6.8	2.5
32-64	(5) 8.5	(12) 5.4	(4) 6.6	(21) 6.3	2.6
Average	(45) 6.9	(156) 6.3	(54) 7.2	(255) 6.6	
Massive bed					
0-2	(5) 3.9	(31) 5.4	(15) 3.3	(51) 4.6	2.0
2-4	(8) 4.0	(24) 4.8	(6) 3.2	(38) 4.4	1.6
4-8	(5) 3.6	(32) 5.0	(9) 3.1	(46) 4.5	1.7
8-16	(11) 4.0	(35) 4.3	(7) 4.3	(53) 4.2	1.8
16-32	(10) 4.3	(29) 4.8	(11) 3.4	(50) 4.4	1.4
32-64	(4) 6.4	(10) 5.3	(5) 2.8	(19) 4.9	1.8
Average	(43) 4.2	(161) 4.9	(53) 3.4	(257) 4.5	

of the parting shale were cut by the Copper Range Co. at 20-foot intervals; each foot was assayed separately. The distribution of grades in practically every channel is such as to permit a very simple division of the total copper into two parts, one representing the lower laminated bed and the first foot of lean rock above it and the second including the copper in the balance of the parting shale. The lean rock just above the laminated bed is a conspicuous element in the assay record of almost all channel samples, and it actually makes very little difference in respect to total copper whether it is included with the upper or lower fraction.

Tables 3 and 4 give the average copper content, in foot-percent, of the lower and upper parts of the parting shale for groups of samples at different perpendicular distances from faults. Data are presented separately for three different parts of the Schacht shaft exploration, as well as for the exploration

as a whole. The unbracketed figures after "0-2" in each table, for example, give the mean copper content in foot-percent of the upper or lower unit in all the channels in the given part of the mine that are within 2 feet of a fault. Figures in parenthesis are the number of channels included in the average. Table 3 gives the data for distance of channel sample from any fault, including some gouge slips with practically no visible displacement; in Table 4 only faults with displacements of more than a foot, horizontal or vertical, are considered.

The differences in copper content between the various categories in Tables 3 and 4 are mostly small, except in the case of categories represented by only a few samples. And in general such differences as do exist are unsystematic.

TABLE 4
COPPER IN PARTING SHALE VERSUS DISTANCE FROM FAULTS WITH MORE THAN A
FOOT OF DISPLACEMENT, SCHACHT SHAFT EXPLORATION

Distance from faults (feet)	Amount of copper (foot-percent) (Number of channels in parenthesis)				Standard deviation from average for exploration
	Northwest part	Middle part	Southeast part	Average for exploration	
Laminated bed					
0-2	(5) 6.3	(15) 6.8	(5) 5.6	(25) 6.5	2.1
2-4	(4) 9.0	(15) 6.3	(4) 6.3	(23) 6.7	2.7
4-8	(4) 8.1	(16) 6.0	(3) 7.0	(23) 6.5	2.4
8-16	(10) 6.7	(27) 5.9	(11) 6.3	(48) 6.1	2.3
16-32	(9) 7.7	(36) 6.4	(12) 7.6	(57) 6.8	2.2
32-64	(9) 7.2	(32) 6.2	(9) 8.5	(50) 6.8	2.6
64-128	(3) 1.4	(15) 6.3	(10) 7.4	(28) 6.2	2.6
Average	(44) 6.9	(156) 6.3	(54) 7.2	(254) 6.6	
Massive bed					
0-2	(4) 3.3	(14) 4.9	(5) 2.7	(23) 4.2	1.7
2-4	(4) 3.8	(14) 5.2	(4) 2.9	(22) 4.5	1.9
4-8	(4) 3.5	(18) 5.0	(3) 3.4	(25) 4.5	1.5
8-16	(10) 3.5	(28) 4.9	(11) 3.4	(49) 4.3	2.0
16-32	(9) 4.3	(36) 4.4	(12) 4.3	(57) 4.4	1.4
32-64	(9) 5.4	(33) 5.1	(8) 3.3	(50) 4.9	1.7
64-128	(3) 5.7	(16) 4.8	(10) 2.6	(29) 4.2	1.5
Average	(43) 4.2	(159) 4.9	(53) 3.4	(255) 4.5	

The only data that suggest any systematic variation are those from the upper part of the parting shale in the northwestern part of the exploration; there is a suggestion, particularly in Table 4, that the amount of copper increases away from faults, though there probably are not enough samples to make the trend very certain. With this possible exception, the data from the Schacht shaft provide no basis whatever for assuming any relation between copper content of the parting shale and faults—they indicate that, at least on a scale of tens of feet, the two are quite independent.

The percentage standard deviation of individual 1-foot samples of the same bed generally averages between 32 and 43 percent, which indicates rather marked variability, due either to sampling technique or to actual varia-

tion from place to place. Some pairs of 1-foot samples cut from the same spot disagreed by as much as 20 percent in a few cases. Such differences may be explained, in part at least, on the basis of accidental inclusion and exclusion of chalcocite nodules and veinlets.

A second method of evaluating the importance of faults and fissures in relation to copper content is through comparison of the copper content of the two parts of the parting shale in individual channels. If the copper content of the upper half of the bed in each channel as ordinate is plotted against the copper content of the lower half of the same channel as abscissa, a scatter diagram results. Enrichment or impoverishment near faults should create a tendency, at least, for the copper content of both parts of the parting shale to be either rich or poor in the same channel; this should lead to a cluster of points that is elongate in a direction away from the origin. A diagram of this sort that was prepared shows only a slight tendency for the copper to be high in one bed where it is low in the other; the low negative correlation coefficient for the two sets of figures, -0.214 , expresses the same suggestion numerically. If there is any actual relationship between the amount of copper in the upper and lower parts of the parting shale, therefore, the relationship is not that which should be produced by enrichment near faults.

One mineralogical observation in the Schacht shaft also suggests a lack of correlation between faults and mineralization. Chalcocite occurs in the upper shale 6 to 8 feet above the upper sandstone in the westernmost part of the drilled area (Zones A and B, Fig. 10D), and pyrite occurs in the same stratigraphic interval farther east. Where the Schacht shaft itself cuts this bed only pyrite is present, but 1,000 feet farther west in an exposure in a drift the bed contains only chalcocite. The pyrite exposed in the shaft is adjacent to a large fault (shown cutting the shaft in Fig. 7), whereas the chalcocite farther west is as much as 80 feet from the nearest fault. If this phenomenon is explained on the basis of normal hydrothermal zoning away from faults, the regional distribution is anomalous; pyrite is more abundant in the upper shale far from the White Pine fault than near it and is found high in the stratigraphic column rather than low. If pyrite is regarded as a sedimentary mineral that is replaced by chalcocite, the copper being hydrothermally introduced, the presence of pyrite next to a large fault is anomalous. The local and regional relationships are consistent only if the mineralogy is unrelated to the faulting.

The influence of tectonic features can be at least tentatively evaluated for the area as a whole. As was shown in the section on structure, the intensity of deformation, as measured by the abundance and magnitude of faults and by the amount of curvature of beds in the White Pine anticline, definitely decreases northeastward away from the White Pine fault. Individual minor faults tend to strike about $N 25^{\circ} W$. If copper content is related to structure, therefore, lines of equal copper content over the area as a whole should tend to show some sort of parallelism with the White Pine fault and anticline. In detail they might show parallelism with minor faults and have northwesterly trends.

The grade of the laminated bed at the base of the parting shale (Fig. 9B)

is richer in a few spots close to the White Pine fault than away from it; the large area with more than 2 percent copper, however, is not parallel to the fault at all but extends directly away from the fault for at least $3\frac{1}{2}$ miles as a belt about $1\frac{1}{2}$ miles wide. It crosses the crest of the White Pine anticline at right angles. Conceivably there may be some enrichment of the bed near the fault; the richness of the copper-bearing area as a whole, however, seems to be quite independent of structural control.

The areal distribution of grades in the laminated bed at the base of the upper shale (Fig. 9D) is far more erratic than that of the bed at the base of the parting shale. There is, however, no apparent close association of high copper content with either the White Pine fault or the White Pine anticline.

The other copper-bearing subzones are less easy to compare from place to place in terms of grade. The maps showing the total copper content of these beds (Figs. 10B and 9D), however, do not suggest any correlation between copper content and faults or folds.

More conclusive, perhaps, than the lack of correlation between structure and copper content in individual beds are the differences in the areal distribution of copper from bed to bed. Structural features like faults and folds should affect all the beds in a given area more or less equally. If, for example, one bed is next to a main feeding fissure in a certain drill hole, all should be, as a general rule. The grades and amounts of copper in the various beds should therefore have a tendency at least to show the same areal patterns of distribution. Perfect correlation cannot be expected, of course, because of differences in thickness and lithology of the host beds from place to place; but the actual correlation is so poor (Figs. 9 and 10) that structural control seems a most unlikely explanation for the present distribution of copper.

In summary, there is no discernible spatial relation of copper in the parting shale to fissures and faults in the Schacht shaft exploration. Similarly, over the drilled area as a whole the distribution of copper in the parting shale and upper shale does not show the complete dependence on structure that is so manifest in the upper and lower sandstones; only the lowermost laminated bed is richer on the southwest flank of the anticline, and much of this superiority may ultimately correlate with the sedimentary features of the beds themselves.

ORIGIN OF COPPER

The copper in siltstone and shale at White Pine is believed to be derived largely from the waters in which the beds themselves were laid down. It is probably more nearly diagenetic than syngenetic, as the chalcocite and some of the native copper may have been precipitated below the surface of the bottom muds or may have replaced primary sulfide; but the copper is believed to have been introduced before the rock was lithified, deeply buried, and deformed. The copper in sandstone on the other hand, particularly the native copper in the old White Pine mine, shows a definite relation to the structure of the area; this is believed to be hydrothermally redistributed copper, possibly derived in large part from the beds at the base of the Nonesuch shale on the

southwest side of the White Pine fault. The basis for these conclusions, largely developed in the foregoing sections, may be summarized as follows:

(1) Copper is found at exactly corresponding stratigraphic positions in the two sedimentary cycles recognized in the cupriferous zone.

(2) The stratigraphic range of high copper content in the individual copper-bearing beds is less than 2 feet, on the average, and is relatively uniform over areas measurable in square miles.

(3) Copper is closely related areally to the depositional environment of the beds that contain it. It occurs in areas that probably had relatively deeper water than their surroundings.

(4) Over the drilled area as a whole the copper does not occur primarily in the permeable sandstone beds. Copper is typically abundant in sandstone or sandy siltstone only within a few inches at most of dark-gray shaly material except locally on the southwestern flank of the White Pine anticline.

(5) In the laminated beds at the base of the parting and upper shales the copper content is less where sandstone laminae are abundant than where such laminae are subordinate. If permeability were important, the opposite should be true.

(6) Copper in massive siltstone in both parting and upper shales is consistently separated by barren rock of similar lithology from the copper-rich laminated beds below. If there were only a single thin cupriferous shale bed overlying a permeable sandstone (e.g., the German Kupferschiefer), one might regard the shale as a membrane that captured the copper, either physically or chemically, from upward coursing waters. But the alternation of copper-bearing and relatively barren beds at the base of the Nonesuch shale is far less readily explained on this basis.

(7) The distribution of copper in the shales and siltstones is not primarily controlled by proximity to the White Pine fault or by the minor faults and the anticline related to it. This is most manifest in the contrast between copper in the sandstone beds, which is largely restricted to the area between the White Pine fault and the anticlinal crest, and copper in the other beds, which shows no such restriction. Only one bed, the lowermost laminated bed, is richer on the southwest flank of the anticline than on the northeast, and it is not at all certain that even this extra richness is due entirely to structural position.

Summarizing the seven preceding arguments, the primary factors controlling the present distribution of copper in the siltstone and shale were stratigraphic and paleogeographic; permeability of the host rocks and structure have had relatively little influence over most of the area drilled by the Copper Range Co. At least three general theories of origin satisfy these conditions:

(1) the copper was dropped into or precipitated within the original mud; (2) the copper, diffusing downward into the uppermost few inches of mud, replaced some primary constituent of the mud, perhaps iron sulfide or calcium carbonate; (3) the copper was introduced into the rocks after lithification and presumably deformation, and the loci of deposition were controlled to an extraordinary degree by the physical or chemical character of the host rocks.

The first two theories require that the copper be derived essentially from the waters in which the beds themselves were deposited, with the copper being deposited only at those times and places at which favorable bottom conditions existed. The third requires an exotic source, presumably in depth.

The first theory, precipitation of the copper on or within the mud of a foul bottom, provides the simplest explanation for the apparent stratigraphic control of its deposition. The second theory, replacement of a primary mineral (e.g., calcium carbonate or iron sulfide) in the foul-bottom mud, is not very different from the first in most respects. It is mentioned primarily because Deans (4, p. 350) has recently suggested that replacement textures in the Kupferschiefer may be primarily diagenetic; although no replacement textures have yet been recognized in preliminary petrographic study of the Nonesuch shale, the numerous points of similarity between the copper deposits in the Nonesuch shale and the Kupferschiefer imply similarity of origin (14).

The third theory requires that the copper be introduced into the rocks after lithification. This possible origin is not incompatible with most features of the stratigraphic distribution of copper if it is assumed that the channels of access have had remarkably little influence on the present distribution of copper, compared with the physical or chemical composition of the host beds themselves. The principal arguments against such an origin are the following:

(1) The most likely channelways for widespread hydrothermal introduction of copper are the more permeable sandstones. In view of the abundant copper found locally in sandstone near the White Pine fault (Fig. 11), it seems unlikely that the sandstones would be so nearly barren over most of the area if they had acted as channelways for all the copper now in shale.

(2) The physical or chemical property that might have controlled the deposition of copper cannot be readily identified. If selective replacement of a primary mineral is invoked, it is difficult to determine what that mineral may have been. Some of the chalcocite now has the form of nodules, and the bedding is bent up and down around these nodules, indicating that they formed before consolidation of the rock. Such nodules might originally have consisted of calcite or pyrite, both of which can still be found as nodules in some beds. But selective replacement of calcium carbonate cannot be invoked because some consistently lean or barren beds between copper-rich beds are everywhere calcareous. If pyrite is the replaced mineral, it is difficult to explain why pyrite in the upper shale next to a large fault in the Schacht shaft is unreplaced. And in those beds of the upper shale that contain pyrite in one place and chalcocite in another, the amount of megascopic pyrite in pyritic places is generally far less than the amount of megascopic chalcocite in other places.

It is equally difficult to determine what physical property of a sediment might have led to such delicate selective precipitation of copper minerals. Copper is found in a variety of rock types, ranging from shale to sandstone.

It is present in both laminated and massive siltstone. It does not typically occur in reddish beds, but on the other hand, it does not occur in all the gray rocks. The rocks above and below the highest copper-bearing bed in the upper shale appear to be lithologically identical with this bed in all respects but copper content.

It may be too early to deny the possibility that the copper was introduced long after deposition of the beds, but in general a theory that requires such delicate selective replacement or precipitation over many square miles, and such independence of control by structure or permeability, seems more strained than a theory holding the copper to be essentially sedimentary in origin.

COPPER IN THE NONESUCH SHALE IN OTHER AREAS

Copper is found in traces, at least, in practically every place where the base of the Nonesuch shale can be observed, from the Bad River near Mellen, Wis., to a point 10 miles southwest of Calumet, Mich., a distance of over 100 miles (Fig. 1). At many points the trace consists only of a little copper stain on joints or bedding planes within a few inches at most of the basal contact; in some of these rocks a few flecks of chalcocite or native copper can be seen. Most of the sections of Figure 6, on the other hand, show chalcocite in quantities that are readily visible megascopically in hand specimen, though the total amount of copper in foot-percent is much less than at White Pine.

Observations at these various localities, each designated by the letter symbols used on Figure 1 and 6, are briefly summarized as follows:

- Locality A. Considerable copper stain in lower 8 to 10 feet and in sandstone just below marker bed.
- B. Considerable chalcocite in sandstone below marker bed near a large fault; sandstone barren away from fault. This copper may represent leakage along the fault from the underlying cupriferous zone, which is not exposed here.
- C. Chalcocite in thin shaly parting 54 feet below marker bed, and in several dark-gray shale beds from 81 to 100 feet below the marker bed.
- D-G. Chalcocite in parting and upper shales as at White Pine but generally much leaner.
- H. Native copper in sandstones as at old White Pine mine.
- I-J. White Pine area.
- K. No copper seen. A little pyrite in sandstone 0 to 1 inch below lowest shale exposed.
- L. A few specks of chalcocite at base of shale.
- M. Base not exposed, no copper minerals seen.
- N. Considerable chalcocite in lowermost few inches of parting shale.

One additional locality should be mentioned. At the White Pine Extension exploration, 5 miles west of Nonesuch mine (H, Fig. 1), 4.8 feet of part-

ing shale averages 1.08 percent copper, and 5.7 feet of upper shale averages 0.6 percent copper (1, p. 59).

The copper content of the cupriferous zone seems to be greatest along the south flank of the Porcupine Mountains in a belt that probably extends west to include much, if not all, of the Presque Isle syncline. South of this belt, where the rocks below the marker bed are sandier (southeast end of section A-A', Fig. 5), the amount of copper is far less; no copper was seen at Gates Creek and almost none at Rockland. North of White Pine, also, in an area with less sandstone (sections D, E, F, G, Fig. 6) that presumably lay farther off shore when the lowermost Nonesuch shale was deposited, the cupriferous zone is appreciably leaner than in the vicinity of White Pine. If the present outcrop of the Nonesuch shale (Fig. 1), excluding its loop around the Porcupine Mountains, is more or less parallel to the old shoreline that existed during Nonesuch deposition, then the belt most favorable for copper must generally lie far down the dip over the region as a whole; in general the facies exposed at the surface is sandy except around the Porcupine uplift. The doming of the Porcupine Mountain area is responsible for bringing this normally deep-lying facies to the level of the present surface in that area. If this facies reaches the surface at any other place, it must be in the area northeast of Greenland, Mich.

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